Approximating Graph Inference by Serving Graph Model Outputs as Views

ABSTRACT

Graph machine learning models are routinely queried for various analytical tasks. Nevertheless, their inference process remain expensive for large graphs. This paper investigates a novel graph inference framework to approximate graph inference by serving historical graph model outputs as "views". Given a graph model M, with its historical outputs over a set of graphs $\mathcal{G}$, and an inference request of M over a large graph G, it packages model outputs from $\mathcal{G}$ along with auxiliary statistics as queryable, memoized view structures called graph model views (GMVs), to directly derive an approximated counterpart of the output of M over G, without an expensive inference process from scratch. Based on GMVs, we study three problems: (1) *Inference using views*, to approximate the output of M by using GMVs with reduced time cost, and when this is possible; (2) *Selection*, to decide whether a set of GMVs can be used to reconstruct the output of M, and if so, which to choose; and (3) *Minimization*, to reduce the storage and access cost of GMVs. For each problem, we establish doability and hardness results, and introduce efficient algorithms for GMV-based inference analysis. Using benchmark tasks, we experimentally verify that GMVs achieves large speedups (850×) with negligible accuracy loss, and scale graph inference to billion-scale graphs. We also showcase the applications of GMV in cybersecurity and bitcoin transaction anomaly detection.

1 INTRODUCTION

Graph (machine learning) models, such as graph neural networks (GNNs) [63] have been routinely queried to support graph analysis. A graph model M converts an input representation of a graph $G = (X, A)$, where X (resp. A) is a matrix of node features (resp. adjacency matrix) of G, to a numerical matrix Z ("logits") via an inference process. Given a graph G, a test node v_t in G, and a GNN M, an "inference query" $Q(M, v_t, G)$ applies the inference process of M over G to return user-specific output $Z(v_t)$ (or Z by default). The output Z can be post-processed for downstream analysis, such as labels for classification, or numerical values for regression analysis. A query workload can be posed to retrieve the outputs $Z(V_T)$ for a set of test nodes V_T, as v_t ranges over V_T in G.

Example 1: Fig. 1 illustrates a fragment of a molecular graph G that encodes a new drug under development. Nodes in G represent atoms and edges represent chemical bonds. It contains $C_6H_8N_2O_4$ (a benzene ring with two hydrogens substituted by functional groups), a useful compound to understand electronic, structural, or reactivity-related properties in computational chemistry [12], However, the presence of its chemically unstable structures such as reactive hydroxylamine groups ($-NOH$) make in-lab tests expensive to verify properties such as polarizability under various settings [61].

A chemical data scientist wants to exploit graph models, *e.g.,* a vanilla GNN M [59]. She may issue an inference query $Q(M, v_a, G)$

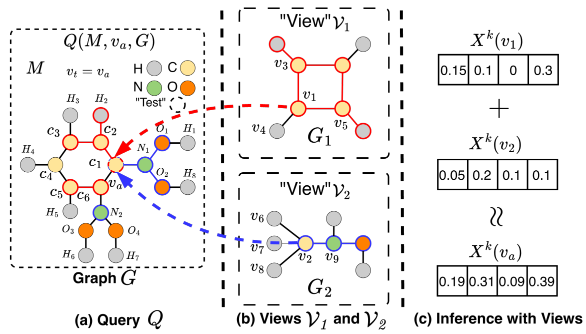


Figure 1: Approximate Evaluation of an Inference Query Q over Target Molecular Graph G Using "Views" $\mathcal{V}_1$ and $\mathcal{V}_2$.

on a designated atom v_a associated with $-NOH$ group to understand how it local chemical structure contributes to predict molecular properties, such as toxicity, solubility, polar surface area, and polarizability, as 4-ary numeric vectors (normalized in $[0, 1]$). By performing node-level inference on v_a, the chemist can estimate the effects of functional groups on the properties. Such output is obtained by an iterative messaging passing over G using an update rule $X_v^k = \sigma(\Theta \cdot \text{AGG}(X_u^{k-1}, \forall u \in \mathcal{N}(v)))$, that requires atom-level features X and an ajacency matrix A of G. After L layers of propagation, M outputs four-dimensional vectors (embeddings) $Z(v)$ at each node v, representing the predicted contribution of atom v to the overall molecular properties. The output $Z(v)$ will be post-processed to derive the aforementioned molecular properties. □

While desirable, such straightforward inference over G remain expensive or infeasible due to access control [16, 45, 49, 54], or when G is large (*e.g.,* a large design space of combinations of compounds). In the above example, data providers may not reveal the entire G to queryers. On the other hand, they may post public data samples at data hub services such as Hugging Face [3], which resemble sufficient "similar" counterpart in G to provide an "approximated" answer to queryers. A natural interest is to best reuse such results as public "views", to approximate $Q(M, V_T, G)$ as fast and accurate as possible, supported by *e.g.,* federated data infrastructures [30].

Example 2: Consider the inference request posed by the chemical scientist. While a direct access to G may undergo long process, metadata is made available on structurally and functionally similar molecules. For example, the chemical properties of Cyclobutadiene (C_4H_4, represented by molecular graph G_1) and methylhydroxylamine (CH_5NO, represented by G_2) are publicly available in chemical databases at ChemSpider [55] and NIST WebBook [44].

Is it possible to approximately answer Q by consulting the model output over G_1 and G_2? While neither are strictly subgraphs of G, they exhibit functional and topological similarities that contribute to similar roles in an inference process over their counterpart in G (*e.g.,* benzene ring-NOH groups). This is justified by fragment modeling in computational chemistry, which verifies that structurally similar fragments demonstrate similar local electronic and structural behavior [7].

We consider wrapping inference outputs over G_1 and G_2 as "views" that can be shared and reused by queryers over a federated infrastructure. Without requesting an inference test of M that require entire G from scratch, the queryer may request an on-site test that derives a "matching" from public G_1 and G_2 to their counterparts that participated inference process in G, such that useful metadata can be shared to approximately reconstruct the output, without revealing G. For example, from the "matching" of v_1 and v_2 to v_a, we can *rewrite* the k-th layer embedding of v_a as:

$$\boxed{\begin{aligned} X^k_{v_a}(\text{in } G) &\approx \sigma(\sum_{u\in N(v_a)} \Theta^k X^{k-1}_u) = \sigma(\Theta^k \cdot (X^{k-1}_{c_2} + X^{k-1}_{c_6} + X^{k-1}_{N_1})) \\ &\approx \sigma(\Theta^k \cdot (X^{k-1}_{v_3} + X^{k-1}_{v_5})) + \sigma(\Theta^k \cdot (X^{k-1}_{v_9})) \\ &\approx X^k_{v_1}(\text{in } G_1) + X^k_{v_2}(\text{in } G_2) \end{aligned}}$$

where X^{k-1}_v (resp. X^k_v) is the embedding of a node v at the $(k-1)$-th (resp. k-th) layer; σ is an activation function, $N(v)$ refers to the neighbors of node v, and Θ refers to the learned weight matrix (same across all layers in the GNN).

This indicates that a queryer only need to acquire metadata derived from on-site matching tests to approximately reconstruct $X^k_{v_a}$, by "rescaling" and assemble the outputs $Z(v'_a)$ from G_1 and G_2. The matching reveals that $Z(v_a)$ has a fraction contributed from a benzene ring structure that is "simulated" via a same inference process at v_1 in G_1; similarly for the other fraction "covered" by the $N(OH)_2$ substituent by rescaling the output at v_2 in G_2. Such metadata (*e.g.,* degrees, weights/attentions, hyper-parameters) can be precomputed and reused in constant time via memoization. □

Better still, we expect an inference time cost bounded by the size of views, independent of the size of G. Analogizing view-based query processing [20], we call this novel strategy *view-based graph inference*. On the other hand, several technical challenges remain.

- How to decide when an inference query Q can be evaluated by such "views" without referring to the original graph G?
- How to reduce the size of views that are needed for such evaluation, hence the computational cost?
- How to approximate original output using views to minimize inference cost with small accuracy loss?

Contribution. We make the following contributions.

(1) We introduce *graph model views* (GMVs) $\mathcal{V}$, a class of memorization structures for view-based graph inference. Likening "views" as queries and their results, a GMV $\mathcal{V}$ packages a "memory" graph $G_{\mathcal{V}}$ (a fraction of published graphs), a memoization structure $\tau_{\mathcal{V}}$ of useful metadata, and the local inference output of M over published $G_{\mathcal{V}}$. The view-based inference aims to approximately answer an inference query $Q(M, v_t, G)$ by only referring to GMVs $\mathcal{V}$.

(2) We study three fundamental problems for view-based graph inference. These analysis provide theoretical justification and algorithm components for view-based inference algorithms.

(a) *View selection.* Given $Q(M, v_t, G)$ and a set of GMVs $\mathcal{V}$, decide whether it suffices to refer to $\mathcal{V}$ alone to answer Q, and identify a minimal subset of $\mathcal{V}'$ that can best serve the need. We provide an efficient selection algorithm with optimality guarantees, quantified by a coverage property that measures how well their matches in G cover the indistinguishable fraction between the memory graphs and the original graph G.

(b) *View minimization.* Given GMVs $\mathcal{V}$, we introduce preprocessing algorithms to obtain a minimal representation $\mathcal{V}_m$ of $\mathcal{V}$ that preserves $Q(M, v_t, G)$ as much as possible. We present two strategies. The first compresses memory graphs into a canonical form by finding nodes that are equivalent in terms of inference process; the second further prunes nodes with contribution dominated by others in terms of inference and evaluation of Q.

(c) *Inference using views.* We introduce efficient algorithms that refers to GMVs $\mathcal{V}$ to approximately answer Q, and provide a sufficient condition when an exact solution is doable. The algorithm scales the result from selected, minimized GMVs. To best reuse GMVs and reduce memory and inference overhead, we introduce a forgetting curve model to keep useful GMVs as long as possible.

(3) We experimentally verify the effectiveness and efficiency of the proposed GMV framework. Our results demonstrate that with *graph-model-as-a-view*, our methods can substantially reduce the inference cost of representative GNNs such as GCNs, GAT, and GraphSAGE by 75.21%-90.53% on average across the studied large-scale graphs, with little to no loss (at most 1.18% and an average of 0.37% loss across all tested large-scale datasets) in inference accuracy. We further showcase the practicality of GMV through case studies in real-world scenarios, including cyberattack detection and anomaly detection in Bitcoin transactions.

Related Work. We categorize related work as follows.

View-based Graph Querying. Several methods have been developed to leverage views for graph pattern query evaluation [20, 41, 62, 71]. [20] proposed an efficient and effective view-based query evaluation for pattern queries. It characterizes what pattern queries can be answered using views by pattern containment. [41] extended the use of views from exact query answering to query approximation and studied subgraph isomorphism query answering using views. It introduces the lower and upper approximation of graph queries using views by considering both simulation and subgraph patterns, based on both graph simulation and subgraph isomorphism. More recently, [62] applied view-based techniques to query evaluations in knowledge graphs. The algorithms in [62] accelerate knowledge discovery process by (1) materializing views as base graphs for summaries to support summarized knowledge search; and (2) developing a practical summarization algorithm as view discover process. [71] developed an automatic view selection in graph databases by proposing an *extended graph view*. The algorithm in [71] ensures that the selected view contains all the edge-induced subgraphs to answer the subgraph and supergraph queries simultaneously.

These prior works focus only on graph query evaluation and do not address the challenge of large-scale graph inference for graph machine learning models. By contrast, (1) we take the first step toward view-based methods to scale large-scale graph inference queries; (2) We propose a "graph model views structure GMVs to encapsulate graph models, test graphs and necessary metadata; and

(3) we design efficient view-based graph inference at scale with a time cost that is independent with graph size.

Accelerating Graph Inference. Existing methods that accelerate GNN inference fall into two categories. (1) Model-centric approaches optimize the model architecture to accelerate inference. Pre-propagation methods such as SIGN [24], PPRGo [10] and SCARA [43] use similar strategy that decouple neighbor aggregation from inference by precomputing multi-hop features. Pruning-based method [75] compresses GNNs by removing redundant parameters in trade of faster inference. While effective, these approaches either modify inference pipeline or rely on model-specific designs, limiting their applicability to pre-trained models and model-agnostic deployment settings. (2) Data-centric approaches aim to accelerate GNNs inference by simplifying the input graph. NeuralSparse [72] adopts a learning-based approach that trains deep neural networks to prune task-irrelevant edges. AdaptiveGCN [40] learns an edge predictor to determine and remove task-irrelevant edges, thereby accelerating the inference process of on CPU/GPU clusters. Such designs are tightly coupled with GCN architecture and lacks model-agnostic applicability. Dspar [47] develops a model agnostic approach, by removing edges with similar “importance” estimated by circuit-inspired resistance measures. SPGC [21] introduces a model-agnostic, inference-friendly graph compression scheme that preserve GNNs inference. A recent work [56] takes similar strategies but retrains graph models from sampled subgraphs with compression strategy.

These methods do not address how to exploit and serve external data resources to scale graph inference with quality guarantees. In our work, we integrate data compression and graph query using views to answer inference queries, marrying the advantages of both tracks. This has not been addressed by prior approaches.

Federated Graph Learning and Inference. Federal graph learning such as FGSSL [35] address structure heterogeneity and non-IID data challenges in local models, and explore data distribution similarities to distill knowledge from a global model into local models. [9, 60] surveyed recent advancements in distributed and parallel graph models. Distributed graph models train GNNs in a distributed cluster by partitioning the workload through data parallelism [6, 13, 36, 73], model parallelism [46, 65, 74], or hybrid strategies [22, 66]. These methods achieve computational efficiency via data or model parallelization. However, they fundamentally differ from our work in two aspects: (1) Our approach focuses on scaling graph inference by best exploiting external graph data and historical inference results, rather than generalizing local models to a global one; (2) GMVs allow memory graphs that are *not* necessarily partitions or subgraphs of the original graph. On the other hand, our algorithms are partition-tolerant, and readily benefit from effective graph parallelization strategies (see Appendix).

The databases community has well-embraced the need for "data-centric native solutions" to support ML/AI applications [4, 76] introduce techniques to improve the scalability and efficiency of ML inference pipelines over graph data. Unlike prior works [21, 40, 47, 72, 72] that focus on designing new GNN architectures to speed up training and inference process, we make a first step to scale GNN inference through the lens of view-based, approximate query processing, advocating model inference outputs to be selectively reused, transformed, and assembled to answer new analytical requests. This enables broader view-based techniques to support large-scale, privacy-aware graph inference.

2 GRAPHS AND GRAPH MODELS

Graphs. A graph $G = (V, E)$ has a set of nodes V and a set of edges $E \subseteq V \times V$. Each node v carries a tuple $T(v)$ of attributes and their values. The size of G, denoted as $|G|$, refers to the total number of its edges, *i.e.,* $|G| = |E|$. Given a node v in G, the *L-hop neighbors* of v, denoted as $N^L(v)$, refers to the set of all the nodes in the L-hop of v in G. The *L-hop subgraph* of a set of nodes $V_s \subseteq V$, denoted as $G^L(V_s)$, is the subgraph induced by the node set $\bigcup_{v_s \in V_s} N^L(v_s)$.

Graph Models. A graph (ML) model $\mathcal{M}$ is a function that takes as input a featurized representation $G = (X, A)$ to an output embedding matrix Z, *i.e.,* $\mathcal{M}(G) = Z$. Here X is a matrix of node features, and A is a (normalized) adjacency matrix of G. [1] In this work, we consider graph neural networks (GNNs) and show that our methods are generally applicable to mainstream GNNs and their specifications.

Graph Inference. We follow a query language perspective [8, 26] to characterize the inference process of a graph model. Given a GNN $\mathcal{M}$ with L layers, each node v is associated with a layer-specific *node update operator* M_v^k at each layer k ($k \in [1, L]$), defined as:

$$M_v^k(\Theta^k, \mathsf{AGG}(X_u^{k-1}, X_v^{k-1}, \forall u \in \mathcal{N}(v)))$$

M_v^k computes X_v^k, which refers to the output node representation at the k-th layer of M ($k \in [1, L]$), determined by (1) the learned model parameters Θ^k, (2) an aggregation function AGG (*e.g.,* $\sum$, CONCAT), and (3) the previous layer embeddings of the neighbors of v that participate in the inference at the k-th layer (denoted as $\mathcal{N}(v)$) and v itself. When $k = 1$, $X_v^0 = X_v \in X$, *i.e.,* the input features.

The *inference process* of a GNN $\mathcal{M}$ with L layers takes as input a graph $G = (X, A)$, and computes the embedding X_v^k for each node $v \in V$ at each layer $k \in [1, L]$, by recursively applying layer-specific M_v^k. Hence, the inference process can be considered as the computation of a composition of *node update operators*:

$$X_v^k = M_v^k \circ M_v^{k-1} \circ \cdots \circ M_v^1(X_v)$$

A GNN $\mathcal{M}$ has a *fixed* inference process, if its node update function is specified by fixed input model parameters Θ, layer numbers L, and aggregator. It has a *deterministic* inference process, if it generates the same output for the same input.

We consider GNNs with fixed, deterministic inference processes. Such GNNs are desired for generating consistent and robust outputs. For simplicity, we assume that M_v specifies a proper set of neighbors that participate in the inference process as $N(v) \subseteq \{u | (u, v) \in E$ or $(v, u) \in E\}$. This formulation allows us to cope with GNNs that exploit neighborhood sampling (such as GraphSAGE), and directed or asymmetric message passing, where the effective neighborhood is determined by the inference rule rather than the raw graph structure. For most representative spatial and message-passing GNNs, the inference process consists of layered aggregation and transformation operations and can be computed in PTIME [15, 75].

[1] A feature vector X_v of a node v can be a word embedding or one-hot encoding [25] of $T(v)$. A is often normalized as $\hat{A} = A + I$, where I is the identity matrix.

GNN Classes. A set of fixed, deterministic GNNs $\mathbb{M}$ belongs to a *class* of GNNs $\mathbb{M}^L$, if for every GNN $\mathcal{M} \in \mathbb{M}$, (1) $\mathcal{M}$ has L layers, and (2) $\mathcal{M}$ uses the same form of node update function M_v^k, for each node $v \in V$ and $k \in [1, L]$. Table 6 (see Appendix) enlists mainstream GNNs with node update functions. Other GNNs are mostly their variants that differs in the choices of aggregation function and other polynomial-time computable operators.

Inference Query. An *inference query* Q is a triple $Q = (M, v_t, G)$ where G is the input test graph, M is a fixed, deterministic GNN model from a class of GNNs $\mathcal{M}^L$ with L layers, and v_t (can be $\emptyset$) is a designated target node. In practice, to answer Q, the model M may provide an API to be invoked, which requiers as input $G = (X, A)$ and perform an inference process to return a task-specific result based on the output-layer embedding $Z(v_t) = \mathbf{x}_{v_t}^{(L)}$.

We define the *output* of Q as (a) the embedding matrix Z, if v_t is not specified *i.e.,* Q=$(M, \emptyset, G)$; or (b) $Z(N^L(v_t)) = \bigcup_{v \in N^L(v_t)} Z(v)$, *i.e.,* a set of embeddings that includes all participating nodes in the inference computation of v_t, which involves up to $N^L(v_t)$ in G, for any GNN $M \in \mathcal{M}^L$ with L layers. The result Z is often referred to as the *logits vectors*, and can be further processed (e.g., via softmax) to produce prediction scores or labels. As the inference process is fixed and deterministic, the output of Q is determined by G and M.

3 VIEW-BASED GRAPH INFERENCE

To enable view-based inference analysis, we introduce a view structure, notably, *graph model views* (GMVs). GMVs are a set of *memoization* structures that record historical output of an inference query $Q(M, v_t, G)$ in a standardized, shareable format.

3.1 Graph Model Views: A Characterization

Given a GNN M, a *graph model view* (GMV) is a triple $\mathcal{V} = (G_{\mathcal{V}}, \mathcal{T}_{\mathcal{V}}, Z_{\mathcal{V}})$, which contains (a) a *memory graph* $G_{\mathcal{V}}$, a graph over which an inference of M is performed, by an inference query $Q(M, v_t, G_{\mathcal{V}})$; (b) a memoization structure $\mathcal{T}_{\mathcal{V}}$, to store the auxiliary metadata (to be discussed); and (c) $Z_{\mathcal{V}}$, the result of Q.

In an analogy of database views, we say a GMV $\mathcal{V}$ is a *virtual view* if it only contains a memory graph, *i.e.,* $\mathcal{V} = (G_{\mathcal{V}}, \emptyset, \emptyset)$, while the structure $\mathcal{T}_i$ and output Z_i are to be derived as needed; Otherwise, it is *extended*, with pre-computed $\mathcal{T}_i$ and Z_i to serve.

Remarks. GMVs are consistently characterized by the semantics of inference queries Q. Two different GMVs may (1) share a same memory graph with different test nodes, (2) involve the same test nodes with different memory graphs, or (3) agree on the test nodes and memory graph, yet bookkeeping different outputs from different GNNs of a class $\mathcal{M}^L$.

View-based Graph Inference. The task of *view-based graph inference* has a general form below (as illustrated in Fig. 2):

- **Input**: A graph G, a GNN $M \in \mathcal{M}$, an inference query $Q(M, v_t, G)$, and a set of GMVs $\mathcal{V}$.
- **Output**: an (approximate) result of Q, computed by referring to $\mathcal{V}$, rather than an inference over G from scratch.

Specifically, (1) if $\mathcal{V}$ are virtual GMVs, one invokes at most $|\mathcal{V}|$ local inference queries over the memory graphs $\mathcal{G}_{\mathcal{V}} = \bigcup G_{\mathcal{V}_i}$; or (2) Otherwise, assemble the extended GMVs (with Z_i, $\mathcal{G}_{\mathcal{V}_i}$ and $\mathcal{T}_{\mathcal{V}_i}$ from each GMV $\mathcal{G}_{\mathcal{V}_i}$ to reconstruct the result for Q.

Algorithm 1 : Algorithm GVInf

Input: An inference query $Q(M, v_t, G)$, a GNN model $M \in \mathcal{M}^L$, relation R^L, a GMV $\mathcal{V}$;
Output: The result of Q.

set $Z := \emptyset$;
/* extension (once-for-all) */
if $\mathcal{V}$ is *virtual* **then**
  set $\mathcal{T}_{\mathcal{V}} := \emptyset$, set $Z_{\mathcal{V}} := \emptyset$;
  Construct $\mathcal{T}_{\mathcal{V}} := \{(v, R^L(v)), S_e\}$ from R^L;
  Compute $Z_{\mathcal{V}}$ by local inference;
/* reconstruction */
for $v \in N^L(v_t)$ **do**
  Retrieve $R^L(v)$ from $\mathcal{T}_{\mathcal{V}}$;
  Initialize $X_v := Z(v')$, where $v' := \text{AGG}(R^L(v))$;
  Retrieve S_u from $\mathcal{T}_{\mathcal{V}}$, $u \in N(v)$;
  Rewrite M_v' with M_v and S_u;
Restore Z with recursive computation using M_v';
return Z;

Figure 2: Procedure GVInf**: inference with a single view.**

Ideally, the result of Q can be exactly reconstructed by *only* referring to $\mathcal{V}$ alone, without visiting G. This is desirable as (1) the overall inference cost is determined by the size of GMVs rather than $|G|$; (2) view-based inference is learning-free and post-hoc, which only require data owners to post GMVs without revealing details of M, raw attribute values, or complete G; (3) the model agnostic algorithms are applicable to any GNN $M \in \mathcal{M}^L$ as long as they adopt a same form of node update function.

3.2 View-based Inference: Doability

We next investigate when the view-based graph inference is doable. We provide a *sufficient* condition that can be checked in PTIME.

General Idea. Bisimulation [2] is a fundamental notion that establishes when two nodes (states) in a labeled graph (transition system) cannot be distinguished by an external observer, and has been exploited as data compression and minimization strategies for graph computation [21]. Given a test node $v_t \in G$, our idea considers graph inference computation for v_t over G as the "observer", and identify the node pairs between G and $G_{\mathcal{V}}$ that cannot be distinguished by the inference process, such that it can be computationally simulated by the inference over $G_{\mathcal{V}}$ directly, leveraging properly stored neighborhood statistics.

We start by introducing a notation of L-bisimulation, a variant of bisimulation that considers embedding equivalence and L-hop neighbors of a designated node. Given two graphs $G_{\mathcal{V}}$ and G, a test node v_t in G, and an integer L, let $G^L(v_t)$ be the L-hop subgraph of v_t in G. An *L-bisimulation at v_t*, denoted as $R_{v_t}^L$, is a binary relation between the nodes of $G_{\mathcal{V}}$ and $G^L(v_t)$, such that

- (1) there exists a node v_t' in $G_{\mathcal{V}}$ where $X_{v_t}^0 = X_{v_t'}^0$; and
- (2) for each pair $(v, v') \in R^L$, (a) $X_v^0 = X_{v'}^0$, and (b) for *every edge* (u, v) in $G^L(v_t)$, there is a *match* (u', v') in $G_{\mathcal{V}}^L(v_t')$, such that $(u, u') \in R^L$; and for *every edge* (u', v') in $G_{\mathcal{V}}^L(v_t')$, there is a match (u, v) in $G^L(v_t)$, such that $(u, u') \in R^L$.

We say v'_t and v_t are L*-bisimilar* if an L-bisimulation between $N^L(v'_t)$ and $N^L(v_t)$ contains (v_t, v'_t). When v_t is specified, we denote $R^L(v_t)$ as R^L for simplicity. We can verify the following properties of L-simulation.

Lemma 1: *Given G with v_t, $G_{\mathcal{V}}$, and integer L, (1) an L-bisimulation R^L is an equivalence relation; and (2) a maximum L-simulation R^L can be computed in $O(|G_{\mathcal{V}}| + |N^L(v'_t)||G^L(v_t)|)$ time.* □

One can verify that L-bisimulation is reflexive, symmetric, and transitive. The maximum L-simulation can be computed by a verification procedure (denoted as L-Bisim, not shown) that (1) verifies the existence of v'_t in $O(|G_{\mathcal{V}}|)$ time, and (2) computes the maximum bisimulation relation between $N^L(v'_t)$ and $N^L(v_t)$. This can be implemented by invoking fast bisimulation algorithms [27, 50], in $O|N^L(v'_t)||G^L(v_t)|)$ time. Here L is in practice small for GNNs.

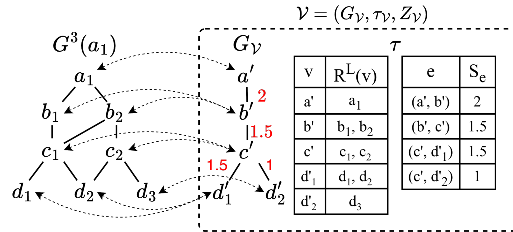
Theorem 2: ***(Doability of View-based Inference).*** *Given a* GNN *class $\mathcal{M}^L$, for any* GNN *$M \in \mathcal{M}^L$ with node update function M_v, an inference query $Q(M, v_t, G)$ can be answered by only referring to a* GMV *$\mathcal{V}$, if there exists a node v'_t in $G_{\mathcal{V}}$ that is L-bisimilar to v_t.* □

We next provide a constructive proof of Theorem 2. Given an inference query $Q(M, v_t, G)$ with M from a GNN class $\mathcal{M}^L$, and a GMV $\mathcal{V}$, assume there is a node v'_t in $G_{\mathcal{V}}$ that is L-bisimilar with v_t (verified by procedure L-Bisim). We clarify (a) the construction of $\mathcal{T}_{\mathcal{V}}$ for GMV $\mathcal{V}$, if $\mathcal{V}$ is virtual; and (b) provides a procedure, denoted as GVInf, to approximately compute $Q(M, v_t, G)$ by referring to the extended GMV $\mathcal{V}$ only. This procedure is used as a "building block" for our general view-based inference algorithms, when the sufficient condition may not hold for a single GMV (see Section 4).

View-based Graph Inference. We outline the procedure GVInf. It takes as input an inference query $Q(M, v_t, G)$, $N^L(v_t)$, $N^L(v'_t)$, and the maximum L-bisimulation R^L computed by procedure L-Bisim, and evaluates Q with two steps below.

(1) **Extension**. If $\mathcal{V}$ is virtual, GVInf performs a "once-for-all" computation to extend $\mathcal{V}$ by constructing $\mathcal{T}_{\mathcal{V}}$ and $Z_{\mathcal{V}}$. (a) Given the node update function M_v, $\mathcal{T}$ is a set of key-value pairs that compactly encodes R^L with the bisimilar node pairs $(v, R^L(v))$, and for each edge e in $N^L(v_t)$, the number of its matches S_e. (b) It then performs a local inference of M over $N^L(v'_t)$ to store $Z_{\mathcal{V}}$.

(2) **Reconstruction**. Given the GNN class $\mathcal{M}^L$ with update node function M_v, it *represents* M_v in terms of a counterpart that directly operates on $R^L(v)$ in $G_{\mathcal{V}}$. (a) For each node v in $N^L(v_t)$, it retrieves $R^L(v)$ from $\mathcal{T}$, which is the set of its L-bisimilar counterparts (an equivalent class), initializes X_v with $Z(v')$ where v' aggregates information from all nodes in $R^L(v)$. (b) We next rewrite each neighbor messages X_u^{k-1} ($u \in N(v)$) in M_v by replacing it with a scaled message from the corresponding representative neighbor $u' \in \mathcal{N}(v')$ in $G_{\mathcal{V}}$. For an edge (u', v') in $G_{\mathcal{V}}$, *scaling factor* $s_{e(u',v')} = \frac{|\{(R^L(u'), R^L(v')) \in N^L(v)\}|}{|R^L(v')|}$ is computed at query-time, where the numerator counts how many matched edges in $N^L(v)$ for edge (u', v') and $|R^L(v')|$ denotes the number of L-bisimilar nodes of v'. The reconstructed message from u' to v' is thus given by $s_{e(u',v')} X_{u'}^{k-1}$ and then determined by determined by the aggregators of M_v (*e.g.,* $\sum$, AVG), the neighborhood statistics of u (such as node degrees, edge attentions, etc). These are all bookkept in



v	$R^L(v)$
a'	a_1
b'	b_1, b_2
c'	c_1, c_2
d'_1	d_1, d_2
d'_2	d_3

e	S_e
(a', b')	2
(b', c')	1.5
(c', d'_1)	1.5
(c', d'_2)	1

Figure 3: A illustration of View-based Inference. The inference for test node a_1 in $G^3(a_1)$ is "simulated" by a counterpart for a' that 3-bisimilar to a_1 in a GMV, for GNN class $\mathcal{M}^3$.

$\mathcal{T}_{\mathcal{V}}$ from $\mathcal{V}$. Such scaling factors can be pre-determined for mainstream GNN classes, or directly included in $\mathcal{T}$ during the once-for-all extension stage. Table 1 summarizes the resulting rewritten node update functions, where the red terms highlight how scaling factors $s_{e(u',v')}$ are combined with degrees information in GCN, attention weights in GAT, and restoration factors in GraphSAGE. This shows that mainstream GNNs can be reconstructed within in a unified edge scaling schema.

Example 3: Consider an inference query $Q(M, a_1, G)$ for a Vanilla GNN $M \in \mathcal{M}^3$ (L=3), and the 3-hop neighbor graph $G^3(a_1)$ as shown in Fig. 3. A GMV $\mathcal{V}$ is initialized with a memory graph $G_{\mathcal{V}}$. The nodes with same labels a, b, c, d have the same input features except $X^0_{d_1} = X^0_{d_2} \neq X^0_{d'_2}$.

(1) In the extension phase, procedure GVInf invokes procedure L-Bisim to verify if a 3-bisimilar exists, and derives the following R^3: $\{(a', a_1), (b', b_1), (b', b_2), (c', c_1), (c', c_2), (d'_1, d_1), (d'_1, d_2), (d'_2, d_3)\}$; and (2) for each edge e in $G^L(a_1)$, at least one match in its match set. It then performs local inference of M over $G_{\mathcal{V}}$ to store $Z_{\mathcal{V}}$.

(2) In the reconstruction phase, GVInf reconstructs the inference over the view graph G_V using $\mathcal{T}_{\mathcal{V}}$ and $Z_{\mathcal{V}}$. For each node $v \in G_3(a_1)$, it first initializes its feature representation as $X_v = Z(v')$ where node v' is aggregated from nodes in $R^3(v)$. It then multiplies each neighbor message X_u^{k-1} in M_v by the corresponding scaling factors retrieved from T_V. Since M is a 3-layer Vanilla GNN, the scaling factors for all edges are derived accordingly and bookkept in $\mathcal{T}_{\mathcal{V}}$. For example, consider edge (b', c'), there are three matches in $G^3(a_1)$ for edge (b', c') and $|R^3(b')| = 2$. Hence, the scaling factor associated with (b', c') is $s_{e(b',c')} = \frac{3}{2}$. Referring to Table 1, it rewrites $X^k_{c_1} = \sigma(\Theta \cdot \mathrm{AGG}(X_u^{k-1}, \forall u \in \mathcal{N}(c_1)))$ as $X^k_{c_1} \approx \sigma(\Theta \cdot \mathrm{AGG}(\frac{3}{2} X_{u'}^{k-1}, \forall u' \in \mathcal{N}(c')))$. All such scaling factors highlighted in red in Fig. 3 are precomputed and stored in $\mathcal{T}_{\mathcal{V}}$, enabling once-for-all reuse. Finally, GVInf recursively applies the updated node function M'_v as illustrated in Table 1 across G_V to reconstruct Z as the inference result of Q.

Now consider a GAT $M \in \mathcal{M}^3$, for the edge (b', c') in the view graph G_V, there are three matched edges (b_1, c_1), (b_2, c_1), and (b_2, c_2) from $G^3(a_1)$ with the learned attention weights $\alpha^{(k)}_{b_1,c_1} = 0.2$, $\alpha^{(k)}_{b_2,c_1} = 0.5$, and $\alpha^{(k)}_{b_2,c_2} = 0.4$, GVInf computes the attention-based scaling factor associated with (b', c') as $s_{e(b',c')} = \frac{0.2+0.5+0.4}{2} = 0.55$. Referring to Table 1, it rewrites $X^k_{c_1} = \sigma(\sum_{u \in \mathcal{N}(c_1)} \alpha_{uc_1} \Theta^k X_u^{k-1})$ as $X^k_{c_1} \approx \sigma(\sum_{u' \in \mathcal{N}(c')} \mathbf{0.55} \Theta^k X_{u'}^{k-1})$. This scaler is stored in $\mathcal{T}_v$ and reused during reconstruction. Compared to the count-based scaling used in Vanilla GNNs, this

GNNs	Node Update Function M_v (at G)	rewriting w. $M_{v'}$; scaling factors are marked in red	notes
Vanilla [59]	$X_v^k = \sigma(\Theta \cdot \text{AGG}(X_u^{k-1}, \forall u \in \mathcal{N}(v)))$	$X_v^k \approx \sigma(\Theta \cdot \text{AGG}(\mathbf{s}_{\mathbf{e}(\mathbf{v'},\mathbf{u'})} X_{u'}^{k-1}, \forall u' \in \mathcal{N}(v')$	AGG: $\sum$ or AVG; for AVG, need to multiple by RF_v
GCN [38]	$X_v^k = \sigma(\Theta^k(\sum_{u \in \mathcal{N}(v)} \frac{1}{\sqrt{deg_u deg_v}} x_u^{k-1}))$	$X_v^k \approx \sigma(\Theta^k(\sum_{u' \in \mathcal{N}(v')} \frac{1}{\sqrt{\mathbf{deg_v}}} \mathbf{s}_{\mathbf{e}(\mathbf{v'},\mathbf{u'})} x_{u'}^{k-1}))$	deg_v: degree of node v in G, topology sensitive
GAT [64]	$X_v^k = \sigma(\sum_{u \in \mathcal{N}(v)} \alpha_{uv} \Theta^k X_u^{k-1}))$	$X_v^k \approx \sigma(\sum_{u' \in \mathcal{N}(v')} \mathbf{s}_{\mathbf{e}(\mathbf{v'},\mathbf{u'})} \Theta^k X_{u'}^{k-1}))$	weight sensitive
GraphSAGE [28]	$X_v^k = \sigma(\Theta^k \cdot (X_v^{k-1} \vert\vert \text{AGG}(X_u^{k-1}, \forall u \in \mathcal{N}(v))))$	$X_v^k \approx \sigma(\Theta^k \cdot (X_{R^L(v)}^{k-1} \vert\vert \text{AGG}(\mathbf{RF_v} \times \mathbf{s}_{\mathbf{e}(\mathbf{v'},\mathbf{u'})} X_{u'}^{k-1}, \forall u' \in \mathcal{N}(v'))))$	$\vert\vert$: concatenation; AGG: AVG
GIN [69]	$X_v^k = \sigma(\text{MLP}((1+\gamma)x_v^{k-1} + \sum_{u \in \mathcal{N}(v)} x_u^{k-1}))$	$X_v^k \approx \sigma(\text{MLP}((1+\gamma)x_{v'}^{k-1} + \sum_{u' \in \mathcal{N}(R^L(v))} \mathbf{s}_{\mathbf{e}(\mathbf{v'},\mathbf{u'})} x_{u'}^{k-1}))$	

Table 1: Rewriting of node update functions for mainstream GNN classes (scaling factors highlighted in red)

attention-aware scaling preserves the relative importance of neighbors induced by GAT's attention mechanism. □

Correctness and Cost. We show that GVInf correctly reconstruct $Q(M, v_t, G)$ from $Q(M, R^L(v_t), G_{\mathcal{V}})$ for a GNN $M \in \mathcal{M}^L$ with an inductive analysis on the inference process of M. The inference executes, at each layer and each node v in $N^L(v_t)$, a node update function to gather the embeddings of all neighbors of v, invoke M_v and update X_v, and sends out the updated embedding to v's neighbors. As there always exists a L-bisimilar node v', such behavior can be correctly simulated by a same process that invokes $M_{v'}$. The only difference is that due to different neighbors, the constant factor that scales X_v differs from X_v'. Such value difference can be eliminated by "re-scale" X_v' with the scaling factor, derived from the recorded edge matches in $\mathcal{T}$. As this computation is consistently simulated at each node $R(v)$, the entire inference process for v_t, which involves only the nodes in $N^L(v)$ in an L-layer GNN M, can be correctly reconstructed by at most L rounds of inference, treated as a composite function that recursively apply M_v up to L times.

For the time cost, observe that GVInf incurs a *once-for-all* overhead in $O(|\mathcal{V}||N^L(v_t')|^2)$ time to extend at most $|\mathcal{V}|$ virtual GMVs, with at most $|\mathcal{V}|$ function calls of the inference of model M. The view-based inference is a direct scaling of recorded embeddings, in total $O(L|N^L(v_t')|)$ time. As L is small ($L \leq 4$), it brings down the inference cost of $Q(M, v_t, G)$ from quadratic-time cost of $|G|$ to a bounded cost only determined by L and $|G_{\mathcal{V}}|$, independent of $|G|$. Specifically, the overhead of computing R^L is relatively much smaller, and on average 3.39 seconds over billion-scale graphs (see Fig. 14, Exp-6 in [1]). This completes the proof of Theorem 2.

The above results can be readily extended to a set of GMVs $\mathcal{V} = \{\mathcal{V}_1, \ldots \mathcal{V}_n\}$. Define the memory graph $\mathcal{G}_{\mathcal{V}}$ of $\mathcal{V}$ as the union of the memory graphs $G_{\mathcal{V}_i}$ of $\mathcal{V}_i \in \mathcal{V}$.

Corollary 3: *An inference query $Q = (M, v_t, G)$ ($M \in \mathcal{M}^L$) can be answered by only referring to a set of* GMVs *$\mathcal{V}$, if there exists a node v_t' in $\mathcal{G}_{\mathcal{V}}$ that is L-bisimilar with v_t in $\mathcal{G}_{\mathcal{V}}$.* □

Example 4: The question in Example 1 can be represented by an inference query $Q(M, v_a, G)$. A set of GMVs $\mathcal{V}$ can be provided, including two virtual GMVs $\mathcal{V}_1$ and $\mathcal{V}_2$, with memory graphs induced from two separately posted sample graphs about Cyclobutadience (C_4H_4) and methylhydroxylamine (CH_5NO), respectively. The extended GMV $\mathcal{V}_1$ sets $\mathcal{T}_{\mathcal{V}_1}$ as two key-value lists that store L-bisimilar node pairs, where keys are nodes in G_1 and values are nodes in the original G. For example, (v_2, C_2) is a L-bisimilar pair. Z_1 is the precomputed result $M(G_1, v_1)$ that represent inference

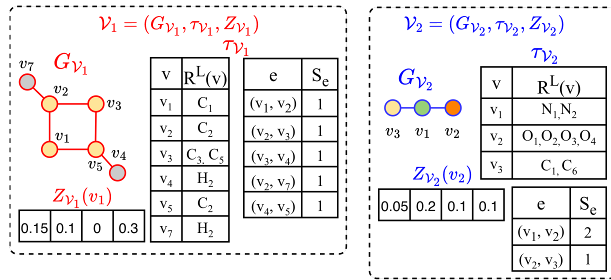


Figure 4: Extended GMVs **$\mathcal{V}_1$ and $\mathcal{V}_2$.**

Algorithm 2 : GVSel

Input: Graph G, inference query $Q(M, v_t, G)$, a set of views $\mathcal{V}$, a test node v_t, maximum number of views k;
Output: Set of views $\mathcal{V}'$ that best covers $Q(G, M, v)$;

```
set 𝒱' := ∅, set U := ∅;
∀𝒱_i ∈ 𝒱, compute C[𝒱_i], /*set of nodes in N^L(v) covered*/;
while ∃𝒱_i ∈ 𝒱 \ 𝒱' : |S[𝒱_i] \ U| > 0 or |𝒱'| < k do
    ∀𝒱_i ∈ V \ 𝒱', S[𝒱_i] := S[𝒱_i] \ U;
    𝒱* := arg max_{𝒱_i ∈ 𝒱\𝒱'} |S[𝒱_i]| where S[𝒱_i] ≠ ∅;
    𝒱' := 𝒱' ∪ {𝒱*};
    U := U ∪ S[𝒱*];
return 𝒱';
```

Figure 5: Algorithm GVSel**.**

result of node v_1 over G_1. Similarly for extended GMV $\mathcal{V}_2$. As there exists a 2-bisimilar relation between the union of the two memory graphs ($\mathcal{G}_{\mathcal{V}}$) and $G^2(a_1)$, one can use $\mathcal{V}$ with scaling factors to approximate $Q(M, a_1, G)$ in Example 1. □

Remarks. In practice, a GMV can be constructed to serve the memory graphs induced as subgraphs from published graph data (via procedure L-Bisim) as needed. For example, the memory graphs $G_{\mathcal{V}_1}$ and $G_{\mathcal{V}_2}$ in Fig. 4 are subgraphs of G_1 and G_2. This ensure the inclusion of GMVs having memory graphs with a 2-bisimilar to G if exists, even when their original counterparts may not.

We next investigate three fundamental problems: view selection, view minimization, and a general task of inference using views when the coverage condition may not hold. This gives a comprehensive treatment of our proposed paradigm, providing provable guarantees to justify its feasibility and doability for large graphs.

4 VIEW SELECTION

For an inference query, one may obtain multiple sample graphs as (virtual) GMVs $\mathcal{V}$ from organizations or data platforms such

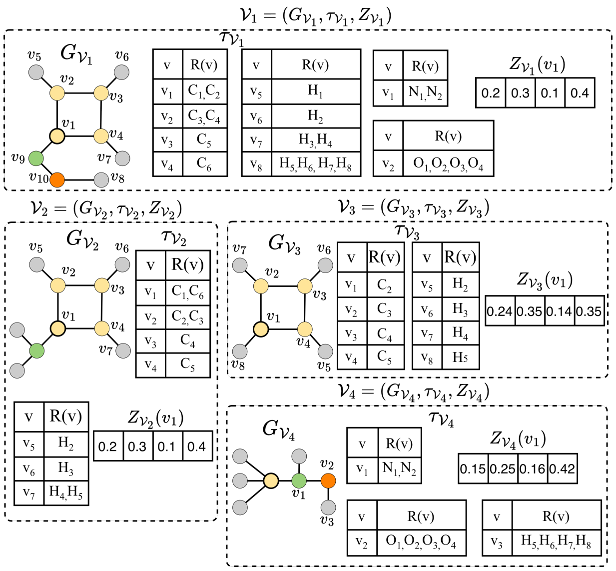


Figure 6: View Selection for Graph Inference. Of the four GMVs $\mathcal{V}_1$, $\mathcal{V}_2$, $\mathcal{V}_3$, **and** $\mathcal{V}_4$, $\mathcal{V}_1$ **completely covers** Q **while** $\mathcal{V}_2$, $\mathcal{V}_3$, **and** $\mathcal{V}_4$ **covers** Q **partially.**

as Hugging Face [3] and OpenML [53]. To ensure fast inference and reduce *e.g.*, the number of function calls of M inference, we investigate view selection problem.

View selection. Given an inference query $Q(M, v_t, G)$, and a set of GMVs $\mathcal{V}$, we want to find a smallest set $\mathcal{V}' \subseteq \mathcal{V}$, such that $\mathcal{G}'_{\mathcal{V}}$ satisfy the sufficient condition (Lemma 3).

This problem is not surprisingly NP-hard: a reduction can be constructed from the set cover problem (see Appendix). We next provide an algorithm, denoted as GVSel, for view selection problem.

Selection strategy. Our algorithm restates view selection by testing a "coverage" property of each GMVs. Given the memory graph $G_{\mathcal{V}}$ of a GMV $\mathcal{V}_i \in \mathcal{V}$, it first computes a non-empty l-bisimilar relation R^l between $G^l_{\mathcal{V}}(v_t)$ and $G^l(v_t)$, for $l \in [0, L]$ (by invoking procedure L-Bisim), and with l the largest possible value (up to L). Here a 0-bisimilar relation R^0 exists, if there is a node v'_t in $G_{\mathcal{V}}$ that has the same feature vector as v_t. For each $\mathcal{V}_i$ that contains a node v'_t that is l-bisimilar with v_t, it sets a "cover" set $S[\mathcal{V}_i]$ as the match set of nodes and edges induced by the l-bisimilar relation R^l.

The algorithm GVSel computes a k-set of GMVs $\mathcal{V}' \subseteq \mathcal{V}$ such that either (1) the union of their memory graphs $\mathcal{G}_{\mathcal{V}}$ satisfy the condition in Lemma 3, or otherwise, (2) it chooses a set of GMVs, such that $\mathcal{G}_{\mathcal{V}}$ covers $G^L(v_t)$ via maximal l-bisimilar relation, hence missing small amount of edges involved in the inference of $Q(M, v_t, G)$, to approximately answer Q. We illustrate the pseudo-code of GVSel in Fig **??**, and present detailed description in [1].

Example 5: Suppose M is a 2-layers vanilla GNN, we illustrate GVSel using the views in Fig. 6. For $\mathcal{V}_1$, all nodes in $N^2(v_t)$ have 2-bisimilar matches in G_1 and all edges preserved in $G_{\mathcal{V}_1}$, so $\mathcal{V}_1$ fully covers Q. For $\mathcal{V}_2$, by the definition of 2-bisimulation, one can verify that $\mathcal{V}_2$ covers all the ten nodes on the benzene ring. For $\mathcal{V}_3$, G_3 contains 2-bisimilar carbon nodes in G, so $\mathcal{V}_3$ covers eight nodes. Finally, for $\mathcal{V}_4$, G_4 simulates all the nodes in the $N(OH)_2$ substituent, giving ten matched nodes. Given $\mathcal{V} = \{\mathcal{V}_2, \mathcal{V}_3, \mathcal{V}_4\}$ and $k = 2$, we demonstrate how GVSel selects views. GVSel computes the set of nodes covered by each view in $\mathcal{V}$. In the first iteration of the while loop, GVSel selects one view from the set $\{\mathcal{V}_2, \mathcal{V}_3, \mathcal{V}_4\}$. GVSel selects $\mathcal{V}_2$ since it covers ten nodes more than the other views. Now, set $U := \{S[\mathcal{V}_2]\}$ and $\mathcal{V}' := \{\mathcal{V}_2\}$. In the second iteration, GVSel selects one view from $\{\mathcal{V}_3, \mathcal{V}_4\}$. GVSel selects $\mathcal{V}_4$ since it covers ten uncovered nodes compared to zero uncovered node by $\mathcal{V}_3$. Now, Set $U := \{S[\mathcal{V}_2], S[\mathcal{V}_4]\} := N^L(v)$ and $\mathcal{V}' := \{\mathcal{V}_2, \mathcal{V}_4\}$. Since $|V'| := k$ and $S[\mathcal{V}_3] \setminus U = \emptyset$ ($\mathcal{V}_3$ cannot be selected since it covers zero node), GVSel returns $\mathcal{V}' := \{\mathcal{V}_2, \mathcal{V}_4\}$. Thus, views $\mathcal{V}_2$ and $\mathcal{V}_4$ are selected by GVSel. □

Analysis. GVSel achieves a $(1 - \frac{1}{e})$-approximation guarantee for computing an optimal set of k views with maximum coverage of $G^L(v_t)$. This follows from a reduction from view selection problem to a cardinality constrained maximum weighted set cover problem. GVSel employs a greedy strategy to select the view that covers the largest number of currently uncovered nodes iteratively, until either budget k is reached and no additional views can be selected. The time cost of GVSel is on $O(k|\mathcal{V}|)$ where $|\mathcal{V}|$ is the number of candidate views. Specifically, it takes $O|\mathcal{V}|$ to compute the set of nodes covered by each view initially. Within each iteration, it takes $O(|\mathcal{V}|)$ to update with uncovered nodes for unchoosen views and select the view that covers the largest number of uncovered nodes. Across at most k iterations, this leads to a total runtime of $O(k|\mathcal{V}|)$.

5 VIEW MINIMIZATION

While view selection allow us to exploit a small set of reusable GMVs that maximize the coverage of $G^L(v_t)$ with their indistinguishable counterparts, treating inference process as an observer, nodes among the memory graphs themselves may contribute to redundancy and a source of inference and storage overhead, especially for large memory graphs.

We next introduce two preprocessing strategies to reduce the size of GMVs. The first compresses memory graphs to smaller representation. The second prunes those that are dominated by other GMVs in terms of coverage ability.

View compression. Our first strategy is to compress $G_{\mathcal{V}}$ with a stronger node equivalence relation that preserves the L-bisimilarity. To this end, We extend inference-friendly compression [21] Given a graph $G_{\mathcal{V}}$, we compute a node equivalence relation R^I between the nodes of $G_{\mathcal{V}}$ and nodes in G, such that for each node pair $(v, v') \in R^I$, (1) v and v' has the same input feature vector, and (2) for any neighbor u of v, there exists a neighbor u' of v', such that $(u, u') \in R^I_{\equiv}$. A compressed graph of $G_{\mathcal{V}}$ is the quotient graph $G^c_{\mathcal{V}}$ induced by R_I, which contains a set of nodes, where each node $[v]$ is an equivalent class of a node in $G_{\mathcal{V}}$; and there exists an edge $([v], [v'])$ if (v, v') is an edge in $G_{\mathcal{V}}$. Let $\mathcal{V}_c$ with $G^c_{\mathcal{V}}$ be a compressed GMV of $\mathcal{V}$, We have the following result.

These view minimization strategies effectively finds redundancy in view-based graph inference. Such redundancy can be mitigated by a "once-for-all" minimization of $\mathcal{V}$ to a counterpart that covers $G^L(v_t)$, hence in turn reduce inference cost as much as possible.

Algorithm 3 : GVMin

Input: A set of GMVs $\mathcal{V}$, inference query $Q(M, v_t, G)$;
Output: A minimized GMV $\mathcal{V}_m$;

1: set $\mathcal{V}_m := (\mathcal{G}_m, \emptyset, \emptyset))$;
2: $\mathcal{G}_m := \bigcup G_i$ (G_i ranges over memory graphs in $\mathcal{V}$);
3: $\mathcal{P} = \{V_m\}$;
4: Refine $\mathcal{P} = \bigcup_{B \in \mathcal{P}}$ Refine$(B, \mathcal{P})$ utill convergence;
5: $\mathcal{G}_m := \mathcal{G}_m / \mathcal{P}$, /*collapse each block into equivalence classes/*;
6: set relation $R^I_\prec := \emptyset$; compute $R^I_\prec$ over $\mathcal{G}_m$;
7: Prune $\mathcal{G}_m$ by removing dominated nodes;
8: Update $\mathcal{V}_m$ with $\mathcal{G}_\mathcal{V}$; update $\mathcal{T}_m$ and Z_m accordingly if not $\emptyset$;
9: **return** $\mathcal{V}_m$;

Figure 7: Algorithm GVMin**.**

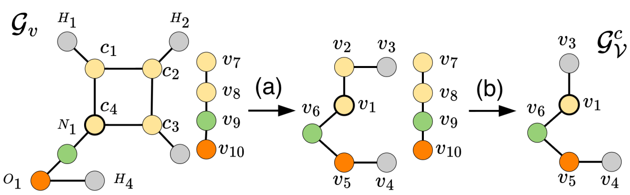


Figure 8: View compression and pruning.

View Dominance. Our second strategy pre-prune GMVs that cannot cover more than another GMVs for inference query $Q(M, v_t, G)$. We extend graph simulation [31] for graph inference, denoted as $R^I_\prec$. Given a graph $G_\mathcal{V}$ with node set V, $R^I_\prec \subseteq V \times V$ is a binary relation such that for any pair $(v, v') \in R^I_\prec$, (1) $X^0_v = X^0_{v'}$, *i.e.,* they have the same feature vector, and (2) for any neighbor u of v in $G_\mathcal{V}$, there exists a neighbor u' of v' in $G'_\mathcal{V}$, such that $(u, u') \in R^I_\prec$. In this case, we say v is inference dominated by v', denoted as $v \prec v'$.

One can verify that $R^I_\prec$ is a preorder (reflexive, transitive), and the maximal relation $R^I_\prec$ can be efficiently computed by a slight revision of the algorithms that computes graph simulation [31] in quadratic time. We observe the following.

Lemma 4: *For any inference query $Q(M, v_t, G)$ that can be answered by referring to $\mathcal{V}$ with $\mathcal{G}_\mathcal{V}$, $Q(M, v_t, G)$ can be answered by $\mathcal{G}_\mathcal{V} \setminus \{v\}$, for any node v that is dominated by another in $\mathcal{G}_\mathcal{V}$.* □

Algorithm. Given a set of (virtual) GMVs $\mathcal{V}$, the view minimization algorithm, denoted as GVMin, preprocesses $\mathcal{V}$ to a minimal representation $\mathcal{V}_m$ with a compressed memory graph $\mathcal{G}^m_\mathcal{V}$ in two phases. (1) The *canonicalization* phase performs a fixpoint, iterative refinement procedure that extends a forward bisimulation [27] with feature equivalence checking. Starting from a coarse partition (line 3), GVMin repeatedly refines blocks by distinguishing nodes with different outgoing-transitions signature until fixpoint stage, *i.e.*,, no further block splits are possible, yielding the coarsest forward bisimulation partition [11, 32] (line 4-6). Each block in the final partition is then collapsed into a node (equivalence class) of $R^I_\equiv$, producing the canonicalized graph $\mathcal{G}^m_\mathcal{V}$ (line 7). (2) The *pruning* phase next computes the maximum inference dominance relation $R^I_\prec$ over $\mathcal{G}^m_\mathcal{V}$ to prune the nodes and edges that are dominated by others, further simplify $\mathcal{G}^m_\mathcal{V}$ to their minimal representation.

Example 6: We next illustrate the view minimization in Fig. 8. Given $\mathcal{G}_v$ with two disconnected parts: 1) the left contains ten nodes and ten edges; and 2) the right part consists of a chain of four nodes, GVMin initializes partition $\mathcal{P}$ to be the set consisting of only one block V_m where all nodes are grouped to one block. It then iteratively refines and splits the blocks based on the structural equivalence. At the end, $\mathcal{P} = \{(c_1, c_2, c_3), (c_4), (H_1, H_2, H_3), (H_4), (o_1), (N_1), (v_7), (v_8), (v_9), (v_{10})\}$. Hence, the left part is reduced into six supernodes v_1, v_2, v_3, v_4, v_5, and v_6 while the right one is untouched. We then continue with dominance pruning. We observe that node v_7 is dominated by v_2 ($v_7 \prec v_2$) since structural coverage of v_7 can be fully simulated by v_2. One can verify that $v_8 \prec v_1$, $v_9 \prec v_6$, and $v_{10} \prec v_5$. If v_1, v_2, v_5, or v_6 is bisimilar to the test node, v_8, v_7, v_{10}, and v_9 can be pruned since v_1, v_2, v_5, or v_6 can provide an equivalent inference counterpart. Conversely, if v_8, v_7, v_{10}, or v_9 is bisimilar to the test node, inference can rely on v_1, v_2, v_5, or v_6 and its neighbors. This shows that dominance-based pruning removes redundancies while preserving full inference coverage. After the compression and pruning, $\mathcal{G}^c_v$ only consists of five nodes and four edges. □

Analysis. The correctness of GMVInf follows from the correctness of procedures GVMin, GVSel and Lemma **??** and Lemma 4. Let $\mathcal{G}^L(v_t)$ of $\mathcal{V}'_m$ contains $|V_m|$ nodes and $|E_m|$ edges. The construction of $\mathcal{G}_\mathcal{V}$ takes $O(k(|V_m| + |E_m|))$ time, and the iterative branching bisimulation partition refinement and inference dominance pruning are in $O(|E_m||V_m|)$. Hence, the total time cost is in $O(k(|V_m| + |E_m|) + |E_m||V_m|)$ per inference query (delay time).

6 APPROXIMATE INFERENCE QUERY WORKLOAD

We next provide a general algorithm, denoted as GMVInf, to approximately evaluate a set of inference queries. This extends our discussion to cope with a set of test nodes V_T.

Algorithm outline. The algorithm GMVInf processes the workload as a stream of inference queries $Q(M, v_t, G)$, as the test node v_t ranges over a set of test nodes in G. We present the pseudo-code of GMVInf in [1]. (1) *Preprocessing*. GMVInf cold-starts by minimizing a set of GMVs $\mathcal{V}$ with the procedure GVMin to compress each of them into their minimal representation respectively. It also invokes the preprocessing component of GVInf to initialize $\mathcal{T}$ and Z, extending any virtual GMVs to extended counterparts, to get it ready to be inferred to directly. This incurs a one-time cost, and the extended GMVs are to be served for the inference workload. (2) *Query Workload Processing*. Upon receiving a query workload W where each inference query $Q_i(M, v_i, G)$ with v_i ranges over a stream of test nodes V_T, it invokes the procedure GVSel to select a set of k views $\mathcal{V}'$ that covers as larger part of $G^L(v_i)$ as possible. If the selected set is not empty, it invokes GVInf to compute the inference results for the corresponding test nodes which approximate the answer of W by rescaling the pre-computed counterparts from $\mathcal{T}$ in the selected GMVs, and append the results to A. Finally, it returns the assembled answer A.

To maximize the reusability of GMVs and further reduce the selection cost, algorithm GMVInf dynamically maintains a cache of memory graph following a forgetting principle [52, 58]. The cache periodically swaps out memory graphs that are no longer "fresh" and retain those expected to be needed, determined by GVSel and a forgetting curve expressed by an exponential decay model. We provide details and an ablation study of its impact to the performance of view-based inference (see Appendix B in [1]).

Analysis. GMVInf correctly computes the approximate answer to query load W as guaranteed by (1) the correctness of GVMin to select the set of top k views $\mathcal{V}'$ that best cover each $q \in Q$ with the greedy strategy; (2) the correct computation of the minimized view $\mathcal{V}_m$ process of GVMin to derive a compact single view representation; and (3) the correctness of GVInf to restore the embeddings of original messaging passing through the scaling factors stored in S_e. We next present time cost analysis of GMVInf. For each query in Q, GVSel is in $O(k|\mathcal{V}|)$ and GVMin is in $O(|E^L(v_t)||V^L(v_t)|)$. For each query in Q, the inference with the single minimized view $\mathcal{V}_m$ is in $O(|G_{\mathcal{V}_m}|)$. The memory graph of the minimized view is bounded by $N^L(v)$. The total cost of GMVInf is in $O(|W|(k|\mathcal{V}| + |E^L(v_t)||V^L(v_t)|))$.

7 EXPERIMENTAL STUDY

Using real-world benchmark datasets, we evaluated GMVInf and its components (view selection GVSel, minimization GVMin, and inference GVInf). We report (1) efficiency (inference cost), and the impact of factors, (2) accuracy of view-based inference, compared with the inference with full access to G, (3) effectiveness of optimization techniques, and (4) cases study in real-world graph analysis. Our source code and datasets are made available[2].

7.1 Experimental Settings

Datasets. We employ seven real-world graph benchmark datasets (summarized in Table. 2). (1) **Cora** [51], a citation network with node features as 0/1-valued word vector indicating the absence/presence of keywords from a dictionary; (2) **Elliptic++** [18] is a Bitcoin transaction dataset, where nodes correspond to wallet addresses and edges represent cryptocurrency transfers, for classifying illicit transactions; (3) **ATLAS** [5], is a cybersecurity log dataset for attack investigation. Each node represents an event (*e.g.,*, system or network activity), and edge denote temporal or causal relationship between events; (4) **Protein** [14], is a molecular graph dataset with 1,113 graphs. Node features include a 3D vector encoding properties such as amino acid type, degree of hydrophobicity, and secondary structure role; (5) **Arxiv** [34], a citation network with nodes representing arXiv papers. Each node has a 128-dimensional feature vector aggregated from word embeddings of its title and abstract; (6) **Yelp** [70] a dense network where nodes represent users and services, and edges refer to friendship. Node features are 300-dimensional word embeddings derived by reviews of services; (7) **Products** [34], a large-scale product co-purchase network.

Besides real-world datasets, we also generated a large synthetic dataset **WS-MAG**, extending the core of the real-world MAG240M citation network [33] using a small-world generator [68]. **WS-MAG** enables controlled studies of the impact of graph size $|G|$ for graphs upto billion-scale. Specifically, we vary $|G|$ from 6 million to 1.194 billion while fixing the average node degree to be 2.

[2]https://github.com/Yangxin666/GMV

Dataset	|V|	|E|	# node types	# attributes
Cora [51]	2,708	5,429	7	1,433
Elliptic++ [18]	5,156	4,784	3	166
ATLAS [5]	59,148	40,823	2	64
Protein [14]	43,471	162,088	2	3
Arxiv [34]	169K	1.2M	40	128
Yelp [70]	717K	7.9M	100	300
Products [34]	2.4M	61.9M	47	100
WS-MAG [33]	0.398B	0.796B	153	768

Table 2: Statistics of Datasets (ranked by graph size $|G|$).

Graph models. We have pre-trained three classes of representative GNNs: GCNs[3] [38], GATs[3] [64], and GraphSAGE[3] [28], for each dataset. For a fair comparison, (1) for all the datasets, we consider the task of node classification, and (2) all methods (when compared against each other) are applied for the same set of GNNs.

Baselines. We compare our algorithm GVInf against the following baselines: (1) GInf, direct inference using original graph G without views; (2) GVInf-all, a variant of GVInf where GVSel refers to all views without forgetting mechanism; (3) GVInf-ran, a variant of GVInf that replaces GVSel with a random selection of k views from $\mathcal{V}$; (4) GVInf-nomin, a variant of GVInf without view minimization. (5) DSpar [47], state-of-the-art graph compression method, performing edge down-sampling to preserve graph spectral information to accelerate GNN inference. (6) SPGC [21] is a graph compression method that builds a compressed G_c where nodes indistinguishable to GNNs are merged. GNN inference is then directly performed on G_c without decompression while preserving inference accuracy.

Evaluation Metrics. We evaluate GVInf, its variants, and baselines with the following measures. (1) Efficiency. (a) Inference Speed-up. We define the speed-up of inference as $\frac{T_{inf}}{T_{vinf}}$, where T_{inf} is the inference time cost of GInf over G, and T_{vinf} represents the inference time cost of view-based and baseline counterparts. Note that T_{vinf} already includes "one-time" preprocessing costs, including view materialization, selection, and minimization.(b) Normalized Compression Ratio (ncr). This measures how well (minimized) views reduce the amount of data to be accessed, compared to the size of L-hops subgraph of the test node. It is defined as ncr $= 1 - \frac{|\mathcal{V}_m|}{|G^L|}$; the closer to 1, the better. (2) Effectiveness. We report the accuracy of GInf and GVInf variants. For Yelp where the benchmark task is a multi-class node classification, we report micro $F1$-score. To ensure a fair evaluation, we report the average performance of 50 different batches of query workload having the same size.

GMVs **Generation and Ground-truth**. We use domain-specific patterns to generate GMVs. For Atlas, we extract frequent cyber attack patterns [5]. For Elliptic++, we use two well-established money-laundering patterns, peel chain and spindles [18]. For Protein interaction dataset, curated functional subgraphs or the entire interaction graph serve as views [14], reflecting chemically grounded structure. For datasets without domain patterns, we reply on frequent pattern mining. In citation networks such as Cora, Arxiv, and WS-MAG, views correspond to frequently occurring citation structures [29]. Similarly, in social and recommendation networks like Yelp and Product, views represent frequently observed recommendation or interaction patterns [39].

	Cora	Proteins	Arxiv	Yelp	Products
GVInf	**5.35 (0.78)**	**41.85 (0.76)**	**80.19 (0.81)**	**138.82 (0.82)**	**849.98 (0.89)**
GVInf-all	4.57 (0.74)	37.48 (0.73)	68.65 (0.78)	114.13 (0.78)	694.81 (0.87)
GVInf-ran	4.66 (0.74)	38.71 (0.74)	64.68 (0.77)	125.41 (0.80)	739.29 (0.88)
GVInf-nomin	2.62 (0.54)	21.53 (0.54)	36.39 (0.59)	61.62 (0.59)	291.01 (0.69)
SPGC	2.23 (0.46)	19.62 (0.49)	31.14 (0.52)	54.67 (0.54)	175.78 (0.49)
Dspar	1.57 (0.24)	15.93 (0.37)	23.31 (0.36)	46.51 (0.46)	124.18 (0.28)

Table 3: Inference Speed-up and Normalized Compression Ratio ncr. ncr **are marked in parentheses. Best performance is highlighted in bold. Inference speed-up is computed relative to** GInf. ncr **is computed relative to $G^L(v_t)$ of test node v_t.**

Given test nodes V_T, and shared "public" subgraphs $G_P \subset G$ induced by $N^L(V_T)$, we construct GMVs $\mathcal{V}$ without accessing to "private" portion $G \setminus G_P$. (1) For each test node $v_i \in V_T$, we extract its L-hops subgraph and extract q subgraphs that matches graph patterns via subgraph isomorphism. By default, $|V_T| = 100$ and $q = 5$, resulting 500 GMVs per dataset. (2) For each $v_i \in V_T$, we sample a set of test nodes V'_T with similar embeddings in the "private" fraction $G \setminus G_P$. The ground truth outputs are generated by inference of M over V'_T. This ensures GMVs with memory graphs having no overlap with private fraction, hence ensure view-based inference to be tested without prior knowledge of the private fraction of G.

7.2 Experimental Results

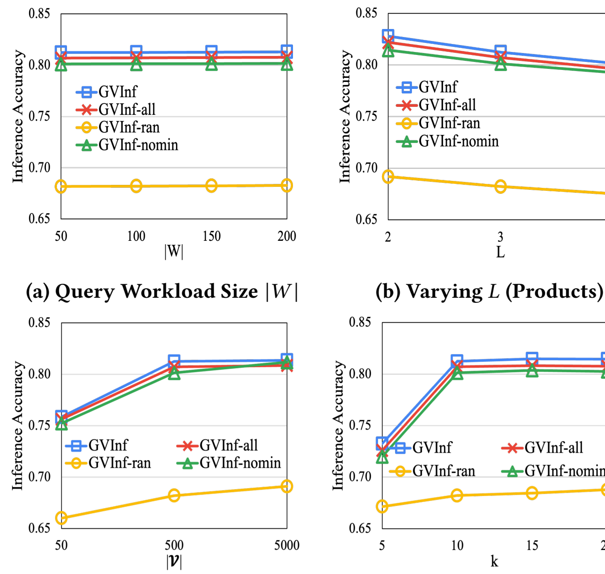
Exp-1: Efficiency of View-based Inference. We first evaluate the time costs of view-based inference.

Inference Speed-up: Overall Performance. Fixing the number of views $|\mathcal{V}|$ being 500, query loads $|W| = 100$, maximum number of views being 10, GNNs model to be a 3-layers GCN, we compare the inference speed-up of GVInf, the variants of GVInf, SPGC, and DSpar over GInf across **Cora**, **Protein**, **Arxiv**, **Yelp**, and **Products** datasets. We report the average inference speed-up and normalized compression ratio (ncr) over all test nodes under a query Q in Table. 3. We observe the followings. (1) GVInf significantly improves graph inference with a speed-up from **×5** to **×850**. The larger G is, the better GMVs-based methods improve GInf. (2) GVInf achieves a greater speed-up in all cases. GVInf-all and GVInf-ran performs better than GVInf-nomin, due to minimization and selection strategies. (3) GVInf and its variants outperform DSpar and SPGC.

Varying Query Workloads Size $|W|$. Fixing $|\mathcal{V}|$ being 500, k being 10, GNNs class to be a 3-layers GCN, we vary $|W|$ from 50 to 200. As illustrated in Fig. 10(a), we find that (1) the inference costs of GVInf and its variants increase linearly as $|W|$ increases; and (2) GVInf achieves the lowest inference costs while GVInf-nomin incurs the highest which is consistent to the observation of the low inference speed up achieved by GVInf-nomin in Table. 3.

Varying Number of Layers L. Fixing $|\mathcal{V}|$= 500, $|W|$= 100, k= 10, we vary the L of GCN from 2 to 4. As shown in Fig. 10(b), the inference cost of GVInf and its variants increase linearly as the number of layers increase from 2 to 4. We remark that this result include once-for-all overhead on extending virtual GMVs, for which we have consistent observation with GNN inference cost analysis in [67].

Varying Number of Views $|\mathcal{V}|$. Fixing $|W|$ being 100, GNN class being a 3-layers GCN, maximum number of views k being 10, we vary



(a) Query Workload Size $|W|$ **(b) Varying L (Products)**

(c) Varying $|\mathcal{V}|$ (Products) **(d) Varying k (Products)**

Figure 9: Inference Accuracy of View-based Inference.

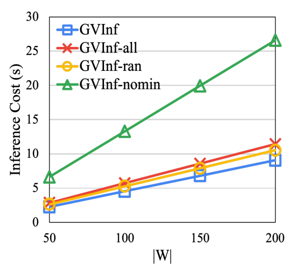
the number of views $|\mathcal{V}|$ from 50 to 5,000. Based on Fig. 10(c), we find that inference cost of GVInf and its variants remain insensitive as $|\mathcal{V}|$ increases. This is because GVInf and variants incur inference cost that is determined by the size of selected GMVs (which is always 10 regardless of how large $|\mathcal{V}|$ is) and GNN classes only, and benefit from fast view selection.

Varying k (# of selected GMVs*)*. Fixing $|Q|$ being 100, the GNN class being a 3-layers GCN, $|\mathcal{V}|$ being 500, we vary the maximum number of views k from 5 to 20 to examine how k affects the inference time cost of GVInf and the variants of GVInf on ***Products***. As shown in Fig. 10(d), when k increases from 5 to 20, the inference costs of GVInf, GVInf-all, and GVInf-nomin all increase first and then gradually level off. This trend occurs because the actual number of selected views actually saturated after k exceeds 10. No more views are selected as k keeps increasing after a certain point. By contrast, inference cost of GVInf-ran increases linearly, as it selects exactly k views, leading to a linear increase in cost with respect to k.

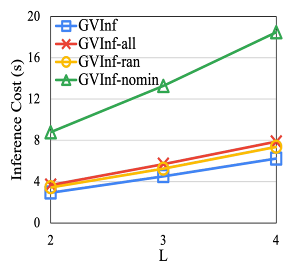
Exp-2: Accuracy of View-based Inference. We next evaluate inference accuracy measured by F1-score (resp. micro F1-score) of single node classification (resp. multi-class node) classification.

	Cora	Protein	Arxiv	Yelp	Products
GVInf	0.7966 ± 0.0767	0.7356 ± 0.0741	0.7148 ± 0.0653	0.6514 ± 0.0712	**0.8031 ± 0.0693**
GVInf-all	**0.8055 ± 0.0604**	0.7323 ± 0.0684	0.7092 ± 0.0614	0.6581 ± 0.0679	0.8017 ± 0.0637
GVInf-ran	0.6722 ± 0.0831	0.6256 ± 0.0815	0.6448 ± 0.0741	0.5531 ± 0.0786	0.6879 ± 0.0752
GVInf-nomin	0.7964 ± 0.0735	0.7302 ± 0.0726	0.7153 ± 0.0656	0.6523 ± 0.0708	0.7985 ± 0.0611
SPGC	0.7784 ± 0.0723	0.7275 ± 0.0741	0.7029 ± 0.0709	0.6342 ± 0.0796	0.7791 ± 0.0764
DSpar	0.7542 ± 0.0794	0.7165 ± 0.0783	0.6875 ± 0.0679	0.6124 ± 0.0773	0.7567 ± 0.0786
GInf	0.7971 ± 0.0121	**0.7387 ± 0.0147**	**0.7175 ± 0.0103**	**0.6592 ± 0.0194**	0.8017 ± 0.0258

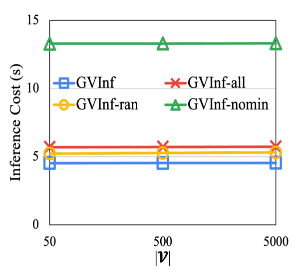
Table 4: Inference Accuracy: Overall Comparison. Best performance is highlighted in bold.



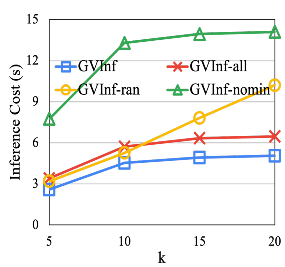
(a) Query Workload Size $|W|$



(b) Varying L (Products)



(c) Varying $|\mathcal{V}|$ (Products)

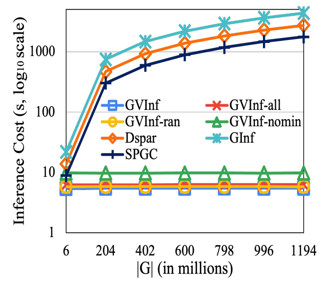


(d) Varying k (Products)

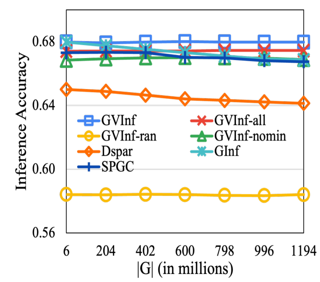
Figure 10: Inference Costs of View-based Inference.

Inference Accuracy. Fixing $|\mathcal{V}|$ being 500, $|Q| = 100$, k being 10, GNNs class to be a 3-layers GAT, we compare the inference accuracy of GVInf and the variants of GVInf across **Cora**, **Protein**, **Arxiv**, **Yelp**, and ***Products*** in Table. 4. We observe the followings: (1) Inference accuracy achieved by GVInf, GVInf-all, GVInf-nomin is generally comparable to each other across all datasets. GVInf, GVInf-all, and GVInf-nomin incurs negligible or no loss relative to GInf across all datasets; and (2) GVInf and its variants outperform DSpar and SPGC, except for GVInf-ran. This result is expected since randomly selected views by GVInf-ran tend to provide poor coverage of Q, leading to reduced inference accuracy.

Varying Query Workloads Size $|W|$. Fixing $|\mathcal{V}|$ being 500, k being 10, GNNs class to be a 3-layers GCN, we vary $|W|$ from 50 to 200 to compare inference accuracy of GVInf and the variants of GVInf over ***Products***. As shown in Fig. 9(a), we observe the followings. First, the inference accuracy of GVInf, GVInf-nomin, and GVInf-all remain stable and comparable to each other as $|W|$ increases. Inference accuracy is insensitive to the number of test nodes since the test nodes are randomly selected. Second, GVInf-ran incurs the



(a) Inference Cost



(b) Inference Accuracy

Figure 11: The Impact of $|G|$ on View based graph inference .

lowest inference accuracy which is consistent to the observation of the low inference accuracy achieved by GVInf-ran in Table. 4 caused by the poor coverage by randomly selected views.

Varying Number of Layers L. Fixing $|\mathcal{V}|$ being 500, k being 10, $|W|$ being 100, we vary L of GCN from 2 to 4 layers to evaluate how the number of layers affect the performance of GVInf and the variants of GVInf over ***Products***. As shown in Fig. 9(b), the inference accuracy of GVInf and its variants drops as the L increases from 2 to 4. This observation is consistent to the oversmoothing issue encountered by GCN where nodes representation starts to look similar and becomes indistinguishable after multiple layers [57].

Varying the Number of Views $|\mathcal{V}|$. Fixing $|W|$ being 100, k being 10, GNN class being a 3-layers GCN, we vary $|\mathcal{V}|$ from 50 to 5,000. As shown in Figure. 9(c), the inference accuracy of GVInf and its variants improves consistently as $|\mathcal{V}|$ increases, but gradually saturates once $|\mathcal{V}|$ exceeds 500. This saturation occurs because, beyond a certain threshold, the top-k views are sufficient to cover most of the inference query Q. Notably, GVInf-ran has much lower inference accuracy compared to GVInf, GVInf-all, and GVInf-nomin.

Varying the Maximum Number of Selected Views k. Fixing the query loads $|W|$ being 100, GNN class being a 3-layers GCN, the total number of views $|\mathcal{V}|$ being 500, we vary maximum number of views k from 5 to 20. Fig. 9(d) verifies the following. (1) As k increases from 5 to 20, the inference accuracy of GVInf, GVInf-all, and GVInf-nomin all increase first and then gradually level off. This is because the coverage of inference query is saturated after k exceeds 10. No more views are selected even k increases. (2) GVInf-ran exhibits minor accuracy improvements, because it strictly selects k views, which slightly enhances but not substantially expands the coverage of inference query Q.

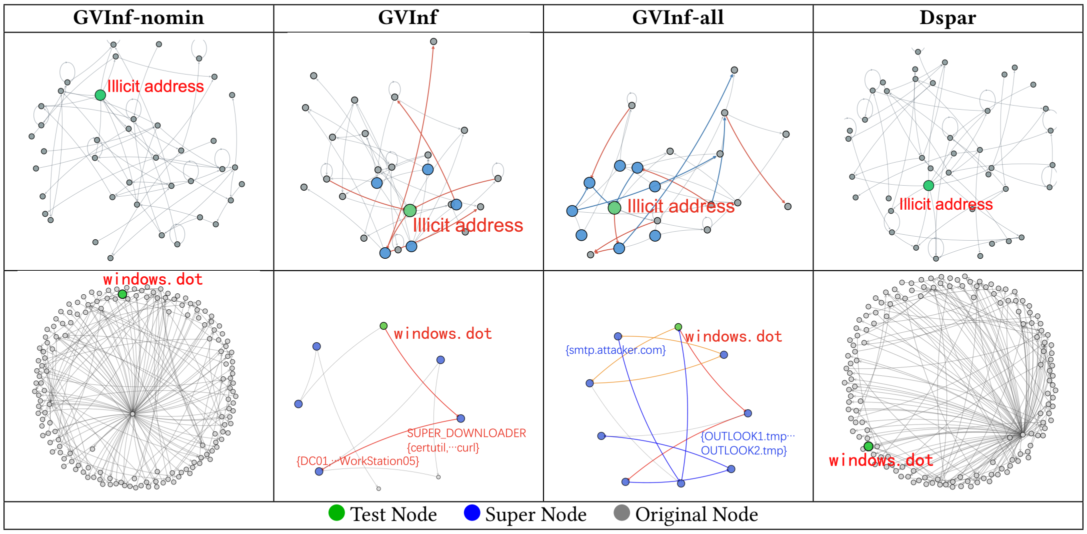


Table 5: Case Study: Bitcoin Laundering and Document Exploit.

Exp-3: Impact of $|G|$. Fixing $|Q|$ being 100, $|\mathcal{V}|$ being 500, k being 10, GNNs class being a 3-layers GAT, utilizing the simulated graph dataset **WS-MAG**, we vary the size of graph $|G|$ from 6 million to 1.194 billion. (1) The inference cost of SPGC, DSpar, and GInf significantly increases (Fig. 11(a)). SPGC induces smaller inference cost than DSpar but still more costly than GVInf and its variants. By contrast, the inference cost of GVInf and its variants remain stable regardless of the graph size, as their inference cost are determined by the size of GMVs $|\mathcal{V}|$, which is a relatively small constant. (2) The accuracy of GInf and its variants remain stable (Fig. 11(b)). Interestingly, the accuracy of DSpar, SPGC and GInf slightly drops. We found that this is due to the output quality of view-based inference also only depends on memory graphs, while DSpar, SPGC, and GInf could suffer accuracy loss due to noisy messages and over-smoothing in large graphs [47, 57].

Exp-4: Ablation Analysis. We also conducted ablation analysis to evaluate the effectiveness of selection, minimization and forgetting mechanism to the accuracy, inference cost, and memory cost, respectively. While each contributed positively to all the measures, GVSel has most significant contribution to accuracy (14.17% loss if removed); and GVMin has contributed most on reducing memory cost and inference cost. We report more details in [1].

Summary. In general, we found that view-based graph inference improve direct inference over large G (at million-billon scale; including **Arxiv**, **yelp**, **Products** and **WS-MAG**) by 488 times on average, accessing only 15.7% of original data, with on average only 0.46% loss of accuracy. These suggest promising application of GMV-based approaches for large-scale graph analysis.

Exp-5: Case Studies. We next showcase use cases of GVInf, in cybersecurity [5] and Bitcoin network analysis [18].

Case Study I: Bitcoin Money Laundering Detection. Anomaly detection in large Bitcoin transaction networks trains GNN-based classifiers to detect whether an account (IP address) is "illicit" or not. We identify two published, small fraction of networks that justify known money laundry patterns such as *Peel Chain* (self-loops) and *Spindle* (disperse–converge) [37], as GMVs (Table 5). (1) All of our GMV-based methods (GVInf-all,GVInf-nomin,GVInf) correctly detected illcit accounts, consistent with the results reported by GInf. (2) With minimization strategies, GVInf achieve smallest cost, involving at most 5 nodes with inference accuracy 0.78, and a ×7 times improvement of GInf. GVInf-all explores larger GMVs with more nodes (ncr **0.78**, inference accuracy **0.81**); and GVInf-nomin involves most nodes in GMVs. DSpar requires a full access of entire network to learn a sparse structure (partly shown).

Case Study II: Attack pattern detection. Cybersecurity analysts train GNNs to predict links in system log graphs to detect attack patterns [5] with malicious access to files (*e.g.,* "windows.dot", Table 5 (Row 2)). Three validated attack patterns are posted and wrapped as GMVs: $\mathcal{V}_1$ (*"Email Delivery":*) (smtp.attacker.com ⟶ outlook.exe ⟶ ··· ⟶ windows.dot), $\mathcal{V}_2$ (*"Exploitation":*) (windows.dot ⟶ dot.exe ⟶ cmd.exe ⟶ certutil ⟶ ··· ⟶ beacon.dll), and $\mathcal{V}_3$ (*"Propagation":*) (windows.dot ⟶ psexec ⟶ net ⟶ DC01 ⟶ ··· ⟶ WorkStation05). While GVInf-nomin explores all three GMVs with an inference over entire 3-hop subnetwork, GVInf can automatically merge equivalent handlers (files):{certutil,wget,curl}:SUPER_DOWNLOADER, yielding **5** supernodes (ncr **0.93**, inference accuracy **0.72**) and incurs smallest inference cost, improving GInf by × **140** times.

8 CONCLUSION

We have proposed graph inference using views, a novel paradigm that scales graph inference by serving and referring to graph model views (GMVs) to approximate the inference outputs, with reduced time and storage costs. GMVs package test data and model output as queryable views. We have introduced fast view selection, minimization, and inference algorithms. Our experiments has verified its efficiency with little loss on inference accuracy. A future topic is to extend GMVs to support inference analysis for dynamic graphs.

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9 APPENDIX

9.1 Appendix A: Notations and Proofs

GNNs **Classes**. In this paper, we focus on mainstream messaging passing-based GNNs such as GCN, GAT, and GraphSAGE. We are aware of other classes of GNNs including spectral GNNs whose node update functions can be approximated by a linear node update function in the form of those used by representative message passing-based GNNs as aforementioned [42]. These linear functions only depend on bounded-hop of neighbors. We will extend our study to these spectral-based GNNs in future.

Proof of Lemma 1. We first show that L-bisimulation at is an equivalence relation over the union of the nodes of $G_{\mathcal{V}}$ and $G^L(v_t)$. (1) It is easy to verify that $(v, v) \in R^L_{v_t}$ (reflexive). To see R^L is symmetric, we show that if $(v, v') \in R^L(v_t)$, then $(v', v) \in R^L(v_t)$. Assume $(v, v') \in R^L(v_t)$, yet $(v', v) \notin R^L(v_t)$. Then either (a) $X^0_{v'} \neq X^0_v$, or (b) there exists a node v'_a that can reach v', or can be reached by v' within L-hops, such that at least for one neighbor u'_a of v'_a on edge (u'_a, v'_a) or (v'_a, u'_a), there is no match of u'_a in $G_{\mathcal{V}}$ to preserve the edge connectivity. Both case contradict to the assumption that $(v, v') \in R^L$. Clearly, for (a), as $X^0_{v'} \neq X^0_v, X^0_v \neq X^0_{v'}$, hence it violates embedding equivalence and $(v, v') \notin R^L$. For (b), if such a u'_a is a direct neighbor of v', then it directly violates condition (2) at test node v_t; otherwise, one can infer that the edge (u'_a, v'_a) has no match in $G^L_{\mathcal{V}}(v'_t)$, which violates condition (2). Hence the contradiction.

To see how R^L can be computed, we outline a procedure L-Bisim, that extends the fast bisimulation algorithms in [27, 50]. (1) L-Bisim verifies the existence of v'_t in $O(|G_{\mathcal{V}}|)$ time, and (2) computes the maximum bisimulation relation between $N^L(v'_t)$ and $N^L(v_t)$, by invoking fast bisimulation algorithms [27, 50], followed by a refinement process that enforces embedding equivalence and iteratively refine by splitting the equivalence classes, and propagate the impact to their neighbors up to L-hop of v_t. This process is in $O|N^L(v'_t)||G^L(v_t)|)$ time. Here L is in practice small for GNNs.

Proof of Corollary 3. Given a set of GMVs $\mathcal{V} = \{\mathcal{V}_1, \ldots, \mathcal{V}_n\}$, we show that GVInf can be readily applied with a preprocessing. It first extends all the virtual views in $\mathcal{V}$ if any. It then creates a single GMV $\mathcal{V}'$ with a single memory graph $\mathcal{G}_{\mathcal{V}}$ as the union of the memory graph $G_{\mathcal{V}_i}$ in each GMV $\mathcal{V}_i$. The processing of Q follows its counterpart GVInf. The only difference is that when updating the inference results, it traces back to the individual $\mathcal{T}_{\mathcal{V}_i}$ and Z_v of the node v in $G_{\mathcal{V}_i}$ to derive the scalars from v's local neighborhood. The correctness follows from the analysis of GVInf.

***L*-bisimulation versus. Bisimulation**. L-bisimulation differs from bisimulation in preserving inference accuracy for GNNs. Classical bisimulation [19, 48] fails to preserve inference accuracy of L-hops GNNs. First, bisimulation defines an equivalence relation over nodes limited to structural indistinguishability of their local neighbors. It is restricted to immediate neighborhoods, which is insufficient for characterizing the full receptive field of an L-hops ($L > 2$) GNNs. Second, bisimulation does not incorporate embedding similarity equivalence. Hence, two bisimilar nodes may induce different inference embeddings after L iterations of messaging passing. In contrast, L-bisimulation enforces equivalence with respect

GNNs Classes	Node Update Function (general form)	Training Cost	Inference Cost
Vanilla [59]	$X_v^k = \sigma(\Theta \cdot \text{AGG}(X_u^{k-1}, \forall u \in \mathcal{N}(v)))$	$O(L\|E\| + L\|V\|)$	$O(L\|E\| + L\|V\|)$
GCN [15, 38]	$X_v^k = \sigma(\Theta^k(\sum_{u\in\mathcal{N}(v)} \frac{1}{\sqrt{d_u d_v}} x_u^{k-1}))$	$O(L\|E\|F + L\|V\|F^2)$	$O(L\|E\|F + L\|V\|F^2)$
GAT [9, 64]	$X_v^k = \sigma(\sum_{u\in\mathcal{N}(v)} \alpha_{uv}\Theta^k X_u^{k-1}))$	$O(L\|E\|dF^2 + L\|V\|F^2)$	$O(L\|E\|dF^2 + L\|V\|F^2)$
GraphSAGE [15, 28]	$X_v^k = \sigma(\Theta^k \cdot (X_v^{k-1}\|\|\text{AGG}(X_u^{k-1}, \forall u \in \mathcal{N}(v))))$	$O(L\|V\|dF + L\|V\|F^2)$	$O(L\|V\|dF + L\|V\|F^2)$
GIN [9, 69]	$X_v^k = \sigma(\text{MLP}((1+\gamma)x_v^{k-1} + \sum_{u\in\mathcal{N}(v)} x_u^{k-1})$	$O(L\|E\|F + L\|V\|F^2)$	$O(L\|E\|F + L\|V\|F^2)$

Table 6: Summary of Representative GNNs. **σ: an activation function *e.g.,* ReLU or LeakyReLU. AGG: aggregation function *e.g.,* sum ($\sum$), average** (Avg), **or concatenation** ($||$). **L, $|E|$, $|V|$, F d: the number of layers, edges, nodes, features per node, and maximum degree.**

to both L-hops neighbors and embedding-level similarity that is necessary for robust inference accuracy preserving.

Hardness of view selection problem. We verify the NP-hardness of view selection problem with a reduction from the set cover problem. An instance of set cover problem (a known NP-complete problem) has a set of elements S, and a set $\mathcal{S}$ of subsets of S. The goal is to find a subset $\mathcal{S}' \subseteq \mathcal{S}$ with size at most k, such that $S = \bigcup_{S_i \in \mathcal{S}'} S_i$. Our reduction is constructed as follows. (1) For each set $S_i \in \mathcal{S}$, We construct a memory graph $G_{\mathcal{V}}$ as a two-level tree with a root v_t and an initial feature vector X_{v_t}. For each element $s_i \in S$, an edge $e_i = (v_t, v_i)$, where v_i is a new node with a unique feature vector assigned to s_i. Similarly, for each subset $S_j \in \mathcal{S}$, a single root tree T_j is constructed, with (a) a root node v_{t_j} with the same feature vector $X_{v_{tj}} = X_{v_t}$, and (b) an edge $e_{jk} = (v_{t_j}, v_{jk})$. We initialize a Vanilla GNN with 2 layers, which simply assigns all the nodes v_{t_j} and the node v_t a label l. Given a cardinality constraint b for view selection problem, and set $b = k$. This construction is in polynomial time. This generates a set of $|\mathcal{S}|$ GMVs $\mathcal{V}$, where each GMV $\mathcal{V} = (T_j, \emptyset, \emptyset)$. We can verify that there exists a set cover of the set S that contains k subsets from $\mathcal{S}$, if and only if a size-k set $\mathcal{V}'$ with a forest $\mathcal{G}_{\mathcal{V}}$ as the union of trees T_j from each GMV $\mathcal{V}_i \in \mathcal{V}'$, that ensures to have at least v_i that is bisimilar to v_t.

Proof of Corollary ??. It suffices to show the following two statements. (1) For any set of GMVs $\mathcal{V}$ with $\mathcal{G}_{\mathcal{V}}$ having a node v_t' that is L-bisimilar to v_t, the canonicalized GMVs $\mathcal{V}_c$ also has a counterpart $[v_t']$ in $\mathcal{G}_[\mathcal{V}]$. This can be shown by observing that for any node pair $(v, v') \in R^L$ and any node pair $(v', v'') \in R_I$ (with v' and v'' in $G_{\mathcal{V}}$), $(v, v'') \in R^L$. (2) To reconstruct $Q(M, v_t, G)$ from the cononicalized GMVs, we only need to update the scalors by a proper calibaration in terms of the neighborhood information over the nodes $[v']$ in $G^c_{\mathcal{V}}$, and the rest processing remain intact.

Proof of Lemma 4. For any nodes and edges participating in the inference computation of v, there exists at least a counterpart in the neighbors of v' that contribute to the computation of v'. If v' is L-bismilar to a test node v_t in an inference query, v contributes no more than v_t in terms of message propagation, hence can be pruned. While if v is L-similar to v_t, then v' always have a counterpart that is L-similar to v_t, induced by $R^I_{\prec}$. Hence, given a set of canonicalized GMVs $\mathcal{V}^c$, we can safely prune all the nodes and their edges that are inference dominated by others via $R^I_{\prec}$.

Notation	Description
$G = (X, A)$	graph G, X: feature matrix, A: adjacency matrix
V_T	test node set
M_v	node update function
AGG	aggregation function in GNN (*e.g.,* $\sum$, Avg, $\|\|$)
x_v^k	embedding of node v at layer k
$\mathcal{M}$, $\mathcal{M}^L$	a GNNs model, set of GNNs models with k layers
$Q = (\mathcal{M}, V_T, G)$	an inference query Q over V_T
$G^L(v_i)$	L-hop induced subgraph of G from v_i
$G^c_{\mathcal{V}}$	minimal memory graph (canonical form)
$\mathcal{V} = (G_{\mathcal{V}}, \mathcal{T}_{\mathcal{V}}, Z_{\mathcal{V}})$	a graph model view $\mathcal{V}$, $G_{\mathcal{V}}$: memory graph, $\mathcal{T}_{\mathcal{V}}$: memoization structure, $Z_{\mathcal{V}}$: the result of Q
R^L, $R^I_{\equiv}$, $R^I_{\prec}$	L-bisimulation relation, inference equivalence relation, inference dominance relation

Table 7: Summary of Notations.

9.2 Appendix B: Algorithms and Analysis

A Parallel Implementation of GVMin. Given sufficient computational resources (*i.e.,* multi-cores CPUs or GPUs), GVMin can be accelerated by adopting the Par-BCRP Algorithm [50], the first known linear-time method for computing strong bisimulation via relational coarsest partitioning. By distributing up to $n = max\{|V^L|, |E^L|\}$ parallel processors, GVMin can parallelize the workloads of iterative branching bisimulation partition refinement across all available units. This ensures GVMin to achieve a linear-time complexity of $O(|V^L| + |E^L|)$ given L and k are small constants, making it highly efficient and scalable for large-scale applications.

Forgetting Model. We also introduce a dynamic maintenance strategy, that aim to keep "useful" GMVs for the query stream as long as possible, hence maximize the reusability of GMVs and reduce the selection cost. Our strategy, inspired by established Ebbinghaus' Forgetting Curve theory [52, 58], poses a forgetting curve model to maintain important memory graphs for GNN M. Following the principle that memory decays exponentially over time, and most forgetting happens soon after learning or inference with a "rate of forgetting" slows down over time (what you remember after some time tends to remain stable longer), We use a view cache to swap out memory graphs that are no longer "fresh" and retain those expected to be needed ($\mathcal{V}_R$), following a forgetting curve expressed by an exponential decay model. Specifically, each view $\mathcal{V}$ is assigned a retention score r:

$$r(\mathcal{V}_i) = e^{-\frac{T_{ep}(\mathcal{V}_i)}{|S(\mathcal{V}_i)|}}$$

where e (the Euler's number) is a constant 2.71828, $S(\mathcal{V}_i)$ refers to the fraction of $N^L(v)$ covered by the memory graph G_i of $\mathcal{V}_i$, and $T_{ep}(\mathcal{V}_i)$ is the time elapsed since $\mathcal{V}_i$ is referred to for evaluating

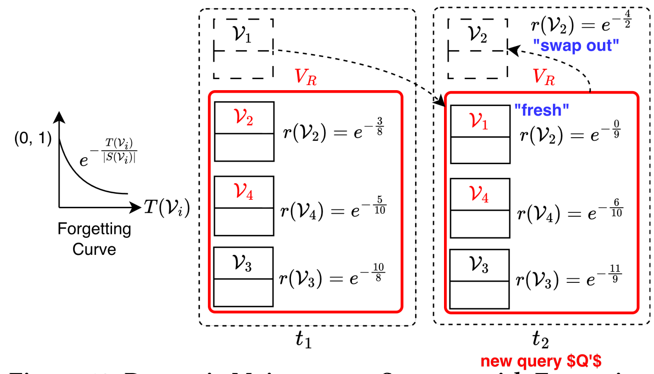

Figure 12: Dynamic Maintenance Strategy with Forgetting Mechanism. Discarding stale views in $\mathcal{V}$ and serves only the current "fresh" subset $\mathcal{V}_R$.

any inference query. In practice, the view cache is dynamically maintained by the procedure GVSel as top $m * k$ views with highest retention scores where $m \geq 1$. The ranking of retention scores ensure that the candidate views in view cache are always the most relevant views to answering Q among the set of all available views $\mathcal{V}$, so GVMin can potentially cover Q with fewer views. Meanwhile, GVMin can also be accelerated since its time complexity is reduced from $O(k|\mathcal{V}|)$ to $O(k|\mathcal{V}_R|)$.

Example 7: Continuing Example 5, two views $\mathcal{V}_2$ and $\mathcal{V}_4$ are selected at t_1. At t_1, $V_R = \{\mathcal{V}_2, \mathcal{V}_4, \mathcal{V}_3\}$. Assume $k = 2$ and cache size $m = 3$, and the time elapse between t_2 and t_1 is equal to 1. Upon receiving a new inference query Q' at t_2, forgetting model updates the retention scores for all the views. For the freshly received view $\mathcal{V}_1$, time elapsed $T_{ep}(\mathcal{V}_1) = 0$ and the size of the fraction covered $|S(\mathcal{V}_1)| = 9$, thus its retention score $r(\mathcal{V}_1) = e^{-\frac{0}{9}}$. For V_4, since $t_2 - t_1 = 1$, the time elapsed $T_{ep}(\mathcal{V}_4) = 5 + 1 = 6$ and $|S(\mathcal{V}_4)| = 10$, $r(\mathcal{V}_4) = e^{-\frac{6}{10}}$. For V_3, $T_{ep}(\mathcal{V}_3) = 10 + 1 = 11$ and $|S(\mathcal{V}_3)| = 9$. For V_2, $T_{ep}(\mathcal{V}_2) = 3 + 1 = 4$ and $|S(\mathcal{V}_2)| = 2$. At t_2, $r(\mathcal{V}_2) = e^{-\frac{4}{2}}$ is smaller than $r(\mathcal{V}_1)$, $r(\mathcal{V}_4)$, and $r(\mathcal{V}_3)$, forgetting model swaps out the view $\mathcal{V}_2$ at t_2 for view $\mathcal{V}_1$. Therefore, at t_2, $V_R = \{\mathcal{V}_1, \mathcal{V}_4, \mathcal{V}_3\}$. Since $k = 2$, only two views $\mathcal{V}_1$ and $\mathcal{V}_4$ are selected at t_2. □

Details of Procedure GVSel. The procedure GVSel first initializes an empty GMV set $\mathcal{V}'$ and an empty coverage set U (line 1). For each candidate view $\mathcal{V}_i \in \mathcal{V}$, GVSel computes $S[\mathcal{V}_i]$ the set of nodes in $N^L(v)$ (line 2). While there are still unselected views that covers uncovered nodes, GVSel conducts the following greedy selection strategy: (1) GVSel updates the uncovered portion of each view by removing nodes already in U (line 4); (2) GVSel selects the view $\mathcal{V}^*$ that covers the largest number of remaining uncovered nodes (line 5); and (3) GVSel adds $\mathcal{V}^*$ to $\mathcal{V}'$ and expand U with the nodes covered by $\mathcal{V}^*$ (line 6-7). This procedure stops when no additional GMVs can be selected.

Query Workload Processing. We present the pseudo-code of the algorithm GMVInf in Fig. 13.

Algorithm 4 : GMVInf (query workload)

Input: Graph G, a set of views $\mathcal{V}$, a set of test nodes V_T, a GNN $M \in \mathbb{M}^L$ with node update function M_v, query workload $W = \{Q_1, Q_2, \ldots, Q_{|V_T|}\}$;
Output: Approximate answer **A** to W ;

set $\mathbf{A} := \emptyset$;
/* preprocessing */
$\mathcal{V}_m := \bigcup_{\mathcal{V}_i \in \mathcal{V}}$ GVMin$(\mathcal{V}_i)$;
/* workload processing */
for $Q_i \in W$ **do**
  /* select a k-subset $\mathcal{V}'_m$ */
  $\mathcal{V}'_m :=$ GVSel $(Q_i, \mathcal{V}_m)$;
  /* view-based inference */
  if $\mathcal{V}'_m \neq \emptyset$ **then**
    $\mathbf{A} := \mathbf{A} \cup$ GVInf$(N^L(v_i), \mathcal{V}'_m, M^L)$;
return **A**;

Figure 13: Algorithm GMVInf**.**

9.3 Appendix C: Complementary Experimental Study and Discussion

Exp-4: Ablation Analysis. Fixing $|W|$ being 500, GNN class being a 3-layers GraphSAGE, $|\mathcal{V}|$ being 500, and k being 10, we conduct ablation studies to evaluate how essential components selection GVSel, view minimization GVMin, and forgetting mechanism FM affects the inference efficiency (including inference time per query load and memory usage of V_m per test node) and inference accuracy (%). We present results computed on average across all datasets as illustrated in Table. 8.

W vs. W/O GVSel. Removing view selection leads to a substantial 14.17% accuracy drop (from 74.24% to 63.72%), 62.03% increases in memory usage per test node and increases inference time by 31.93% from 5.92 s to 7.81 s per query load. This verifies that GVSel effectively prioritizes informative views and reduces redundant information, leading to improved accuracy and efficiency.

W vs. W/O GVMin. Without GVMin, the inference accuracy drops slightly (from 74.24% to 73.21%) while leading to 135.64% increases in inference time from 5.92 s to 13.95 s per query load and 175.32% increases in memory usage per test node. This indicates that the effectiveness of GVMin serve as a computational optimization tool to accelerate inference and reduce the memory footprints without harming inference accuracy.

_W vs. W/O FM_. Disabling the forgetting mechanism, the inference accuracy remains nearly unchanged. However, the inference time rises up 7.09% from 5.92 to 6.34 s per query load and memory usage per test node increases by 36.08%. This demonstrates that the FM component primarily functions as a lightweight optimization strategy to accelerate inference by disregarding stale or less relevant cached views while preserving inference accuracy.

Exp-5: Memory Costs. We conduct memory cost analysis on ***Arxiv***, **Yelp**, and ***Ogbn-Products*** datasets. For each dataset, we report the memory costs of the original graph G and the generated 500 views, including the memory cost of all the memory graphs and all the auxiliary structure respectively. We define memory

Component	Accuracy (%)	Inf. ($s/|Q|$)	Mem. (MB/v)
GVInf	74.24%	5.92	1.58
w/o GVSel	63.72% (↓ 14.17%)	7.81 (↑ 31.93%)	2.56 (↑ 62.03%)
w/o GVMin	73.21% (↓ 1.39%)	13.95 (↑ 135.64%)	4.35 (↑ 175.32%)
w/o FM	74.03% (↓ 0.28%)	6.34 (↑ 7.09%)	2.15 (↑ 36.08%)

Table 8: Ablation study to evaluate key components.

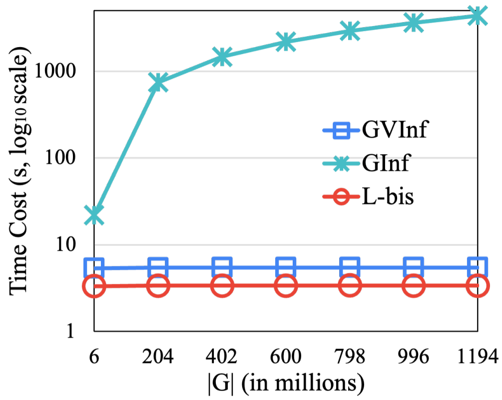

Figure 14: Time Costs of L-bisimulation, GVInf**, and** GInf**.**

compression rate (mcr) as mcr = $1 - \frac{|M_c|}{|M|}$. It quantifies the fraction of memory cost that is "reduced": the larger, the better; The detailed results are presented in Table 9. We conduct the following tests:

(1) **Graph Memory (GM) Test**: We measure the graph memory usage as the graph size in the PyTorch Geometric Graph Data format [23] including node features, node labels, and edge index tensors. Comparing to G, the GM for memory graphs much smaller, reduced by 75.30%, 84.32%, and 85.76% compared to the original graph G for ***Arxiv***, **Yelp**, and ***Ogbn-Products*** respectively;

(2) **Auxiliary Structure $\mathcal{T}$ Memory (ASM) Test**: We evaluate the memory cost of the auxiliary structure $\mathcal{T}$. We observe the followings: i): On average, the memory cost of $\mathcal{T}$ accounts for only 12.13% of that of G_c on average across all the three datasets; and ii) The combined memory cost of $G_c + \mathcal{T}$ still remains significantly smaller than that of G, with reductions of 72.52%, 83.19%, and 83.54% for ***Arxiv***, **Yelp**, and ***Ogbn-Products*** respectively;

(3) **Peak Memory (PM) Test**: We measure the peak memory usage during the inference process using a 3-layers GCN model, using a hidden size of 32. Compared to inference over the original graph G, inference over the set of views $\mathcal{V}$ reduces the peak memory by 65.97%, 81.61%, and 82.60% for ***Arxiv***, **Yelp**, and ***Ogbn-Products***. Notably, as the size of original graph increases, the reduction in peak memory becomes more pronounced, reflecting the overall decrease in both graph size and memory footprints.

Exp-6: Time Costs of L-bisimulation. The cost of full GNN inference which is in $O(L|E|F+L|V|F^2)$, with F reaching millions [17, 34]. In contrast, L-bisimulation is in $O(|N_L(v_t')|\,|G_L(v_t)|)$ time as proved in Sec. 3.2, both are small constants, making it far cheaper. We report the time cost of computing L-bisimulation. Fixing $|Q|$ being 100, $|\mathcal{V}|$ being 500, k being 10, GNNs class being a 3-layers GAT, utilizing the simulated graph dataset **WS-MAG**, we vary the size of graph $|G|$ from 6 million to 1.194 billion to study how graph size affects the time cost of computing L-bisimulation, compared to inference time cost of GVInf and GInf. As $|G|$ increases from 6 million to 1.194 billion, the inference costs of computing L-simulation remains stable regardless of the graph size, as their costs are determined by the size of GMVs $|\mathcal{V}|$ while being significantly faster than GVInf and GInf. On average, it takes 3.39 seconds to compute L-bisimulation over graph with the size of 1.2 billion.

Arxiv	$\|V\|$+$\|E\|$	GM	ASM	PM
G	1,335,586	101.13 MB	–	1.44 GB
$\mathcal{V}$	431,502	24.98 MB	2.81 MB	0.49 GB
mcr	–	75.30%	–	65.97%
Yelp	$\|V\|$+$\|E\|$	GM	ASM	PM
G	14,671,666	1036.03 MB	–	14.41 GB
$\mathcal{V}$	2,742,325	162.51 MB	11.56 MB	2.65 GB
mcr	–	84.32%	–	81.61%
Ogbn-Products	$\|V\|$+$\|E\|$	GM	ASM	PM
G	64,308,169	1887.47 MB	–	30.12 GB
$\mathcal{V}$	11,105,726	268.75 MB	41.78 MB	5.24 GB
mcr	–	85.76%	–	82.60%

Table 9: Memory cost comparison between the original graph G and the set of views graph $\mathcal{V}$.